

Substrate Undecidability

Why No Internal Observer Can Decide Whether Reality Is Discrete or Continuous,
and What Can Adjudicate Instead

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Abstract

We show that the question whether physical reality is discrete or continuous, on its two principal readings (finite vs. infinite information density; computable vs. non-computable dynamics), cannot be decided by any finite-resource observer internal to the system. Two independent barriers are proved and machine-checked in Lean 4: a *topological* barrier — substrate discreteness is not a finitely determined (clopen) property of observable transcripts, so no finite body of evidence decides it (`Substrate.substrate_not_finitely_determined`) — and a *recursion-theoretic* barrier — no candidate law, even when supplied, can be internally certified as the law, by reduction to the halting problem (`Substrate.law_match_undecidable`). Under Perfect Self-Containment (PSC) these results make substrate discreteness an *outsourced* property (`Substrate.substrate_not_internally_decidable`), for which the only admissible adjudicator — by a uniqueness argument this paper cites from the companion NEMS programme rather than re-derives — is minimum-description-length (MDL) selection over complete law-sets, which selects the finite-*description* class: computable transcripts together with an adjudicative extension (“glimmer”) that resolves an apparent tension with the programme’s own account of physical randomness, since the purely computable class alone would contradict its determinism no-go result. We explain why Planck-scale grid searches are structurally unable to settle the question, identify asymptotic scaling exponents as the only finite observables carrying substrate information, and state the results’ limits: the glimmer extension’s classification is fully machine-checked; two physical premises underlying it (that the actual Born measure constitutes complete internal semantics, and that a degree-0’ adjudication bound proved elsewhere for a specific physical framework transfers to this paper’s framework-agnostic model) and the question of which substrate actually obtains are left, honestly, open.

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Code/Data availability: `substrate-undecidability-lean` Lean 4 library (Mathlib only); `lake build`; repository README and module index at repository root give the full axiom audit.

1 Introduction

“Is the universe discrete or continuous?” sounds like one question with a definite, if currently unknown, answer. It is not. It is at least three questions, and they are usually run together:

- (A) **Information density.** Does a bounded region of space admit only finitely many distinguishable states, or infinitely many?
- (B) **Dynamics.** Is the time-evolution law computable (Church–Turing), or does it access non-computable structure (hypercomputation)?
- (C) **Chance.** Is the outcome of a measurement determined, or is there objective randomness?

This paper’s central claim is that neither reading of the question admits an internal answer: no observer that is itself part of the physical system it is trying to characterize, and that has only finite computational resources, can decide axis (A) or axis (B) — not from evidence, and not even if handed a candidate law outright. Both halves of that claim are proved and machine-checked in Lean 4 against Mathlib below (section 5), using nothing beyond ordinary topology and computability theory.

A further question then arises: once no law can be internally certified, is some class of laws nonetheless the one an internal observer should treat as evidentially favored? Answering that second question needs one more ingredient the first does not — *Perfect Self-Containment* (PSC), the standing closure assumption of the broader NEMS research programme this paper draws on. PSC is not yet common vocabulary outside that programme, and the second half of this paper’s argument (Part II, section 8) is built directly on it, so before going any further we ground PSC carefully for a physics audience; that grounding is the rest of this section.

1.1 Perfect Self-Containment, for physicists

Throughout, *Perfect Self-Containment* (PSC) is a criterion on theories, not a hypothesis about the cosmos. A theory satisfies PSC when its predictions are fixed by its own structure: it needs no measure it does not supply, no vacuum selector, no external observer map, and no free completion data to say which of its observationally inequivalent realizations is actual. The NEMS programme proves a trichotomy — any candidate fundamental theory is record-level categorical, internally selecting, or not fundamental — and a reduction showing that every “outside dependency” is a selection in disguise [Spivack, 2025b,p]. Physicists already enforce this condition without naming it: the measure problem of eternal inflation is a problem because the measure is an external chooser [Guth, 2007, Freivogel, 2011]; Wheeler’s “law without law,” Einstein’s Machian objection to absolute space (something that acts but is not acted upon), Everett’s universal wavefunction, the no-boundary proposal, and relational quantum mechanics all exist to remove an unexplained external ingredient [Wheeler, 1983, Einstein, 1916, III, 1957, Hartle and Hawking, 1983, Rovelli, 1996]; and the demand for no free parameters is PSC applied to constants, with minimum-description-length selection [Rissanen, 1978, Solomonoff, 1964] as its formal version — the same adjudication principle this paper cites for Theorem 5 (section 8). PSC is the closure condition of which these are instances.

Its logical status is that of an axiom, and necessarily so: a theory that derived its own closure from a prior principle would make that principle the external selector it had just excluded, in the same way that no consistent system certifies its own consistency [Spivack, 2025k]. What PSC has instead is a self-consistency argument that most axioms lack. To deny PSC is to assert an external selector S . Either S is explained, in which case it belongs to the theory and the theory is internally

selecting — PSC holds; or S is unexplained, in which case the theory is not foundational [Spivack, 2025p]; and any further theory that explains S faces the same dichotomy, so the regress terminates only at an internally selecting stage [Spivack, 2025g,h]. The denial of PSC thus posits exactly what it declares cannot be posited. The programme’s companion note on the PSC–Born fixed point states precisely what is proved and what is assumed [Spivack, 2025a]; the strongest existence evidence is the machine-checked equivalence between PSC and the Born rule as complete internal semantics [Spivack, 2025e,f], which is a fixed point rather than a derivation from nothing. Nothing in Part I of this paper (section 5) depends on PSC; Part II (section 8) does, and section 8 flags exactly where.

A closely related but genuinely distinct question is often folded into axis (B) in informal discussion: whether physical dynamics can solve computationally hard problems *efficiently* — whether nature’s time-evolution lands outside a complexity class such as P or BQP using only polynomial physical resources. Call this **axis (B’): efficient computability**, the complexity class of physical dynamics. Axis (B’) is *not* axis (B): a polynomial-time solver for an NP-hard problem is still a *computable* process — P and NP both sit inside the decidable class — so evidence bearing on axis (B’), efficiency, says nothing about whether dynamics is computable at all, and in particular no such evidence lands anything in **Cstrict** (section 4), which requires genuine non-computability, not mere intractability. The theorems below address axis (B) only; axis (B’) is a complexity-theoretic question tied to the efficient/extended Church–Turing thesis, entirely outside their scope. We flag every place below where evidence properly belongs to axis (B’) rather than axis (B), since the two are conflated often enough in the literature that keeping them apart requires active bookkeeping, not just a definition stated once.

This paper proves, and machine-checks in Lean 4 against Mathlib, that axes (A) and (B) — taken together as “is the substrate discrete or continuum” — cannot be decided by any observer that is itself part of the system being observed and that has only finite resources. This holds in two independent, mutually reinforcing ways. First, no finite body of observed data can decide it: substrate discreteness is not a *finitely determined* property of what an observer sees, in the precise sense that no prefix of observations, however long, settles it (theorem 2). Second, even supplying the observer with a complete candidate law does not help: no internal, finite-resource procedure can certify that the law is the correct one, on pain of deciding the halting problem (theorem 4). Axis (C) is orthogonal to (A) and (B); it is addressed directly by the glimmer corollary (section 8.1), whose classification is machine-checked, leaving the Born-martingale physical premise and a second, distinct framework-transfer premise open (section 10).

The two barriers are of different character, and keeping them apart is one of the paper’s main points. The first — call it the *evidential* barrier — is purely topological: it is a statement about density in Baire space ($\mathbb{N} \rightarrow \mathbb{N}$ with the product topology) and does not use computability at all; consequently no oracle, at any level of the arithmetical hierarchy, changes a density or cardinality fact about Baire space, and nothing escapes this barrier in the oracle sense at all. The second — the *law-based* barrier — is recursion-theoretic: it is a diagonal argument by reduction from the halting problem, and it depends essentially on the observer being computable. Exactly how much extra computational power defeats it is more subtle than “any halting oracle”: $\text{LawMatches}(f, e)$ — “ e matches f at every time step” — is a Π_2^0 statement (a universal claim, over every t , that e halts on input t and agrees with $f(t)$), and LawMatches is in fact Π_2^0 -complete: the reduction proving theorem 4 gives co-halting-hardness, and a second, independent reduction (`Substrate.law_match_pi2_hard`) many-one reduces the totality problem Tot to it, which — since Tot is Π_2^0 -complete, a standard recursion-theoretic fact — pins its complexity exactly at the Π_2^0 level, strictly above the co-halting lower bound alone. This makes the oracle-escape analysis exact rather than one-sided: a $\mathbf{0}''$ oracle (deciding the halting problem relative to $\mathbf{0}'$, sufficient for any Π_2^0 statement) escapes theorem 4; a plain halting ($\mathbf{0}'$) oracle provably does *not*, since no Π_2^0 -complete problem is decidable

relative to $\mathbf{0}'$. This separation (section 6, item 5) is, as far as we are aware, not previously stated in this form, and it matters because the two barriers rule out the two different escape routes one might otherwise propose: “just collect more data” fails against theorem 2; “but suppose the true law were simply handed to us” fails against theorem 4.

Why does this matter, beyond formal tidiness? Because the standard empirical strategy for probing substrate structure — look for a discreteness artifact at some very small scale, e.g. Lorentz-violating dispersion at the Planck length — is not merely difficult. It is attacking a question of the wrong shape. theorem 2 says that *no* finite experiment, run at any scale, with any precision, can settle discreteness in general; and the causal-set programme (section 2) demonstrates concretely, for one specific discretization scheme, that a lattice can be built with no preferred-frame artifact at all [Bombelli et al., 2009]. The lesson generalizes beyond that one scheme: grid searches test whether a *particular* discretization breaks a *particular* symmetry, not whether the substrate is discrete. section 7 identifies what *can* carry substrate information under these barriers: not thresholds, but asymptotic scaling exponents.

Under Perfect Self-Containment (PSC) — the criterion of section 1.1 — the two barriers combine into a structural diagnosis: substrate discreteness is an *outsourced* property in the sense of the Internality/Outsourcing Schema [Spivack, 2025p]. It cannot be decided from inside by evidence or by law-certification, so if it is decided at all, the decision procedure must live outside the space of internal tests altogether. section 8 argues, citing rather than re-deriving the uniqueness argument [Spivack, 2025l,q], that minimum-description-length (MDL) selection over complete law-sets is the unique adjudicator consistent with PSC, and that it selects the finite-*description* class — computable transcripts together with the adjudicative extension formalized in section 8.1 — which is the correct epistemic grounding for treating a compressive, finitely-described physical theory as evidentially favored without ever claiming it has been *certified*.

What this paper is not. It is not a digital-physics paper. It does not argue that the universe *is* discrete, nor does it side with Zuse, Fredkin, Wolfram, or ’t Hooft against a continuum view, nor with the opposite camp. It proves an undecidability result that is agnostic about which substrate obtains, and it explains why every position in that debate is structurally unprovable from the inside — a claim that is stronger than, and different in kind from, taking a side (section 9).

Structure. section 1.1 grounds Perfect Self-Containment for a physics audience, before it is invoked later in this section and in Part II. section 2 traces the intellectual history of this question, from the classical continuum assumption through computability theory to the modern empirical programmes, and situates this paper’s result relative to all of them. section 3 states the three axes and the existing physical evidence bearing on each. section 4 gives the formal model (worlds, transcripts, internal tests, finite determination), matching `Substrate/Basic.lean`. section 5 states and sketches the proofs of the Part I theorems. section 6 states carefully what the theorems do *not* say. section 7 locates the only finite observables that carry substrate information. section 8 states Theorem 5 (outsourcing and MDL) and, in section 8.1, the glimmer corollary that settles axis C’s classification. section 9 is a synthesizing discussion. section 10 lists the remaining open problems, including the Born-martingale lemma for axis C. section 11 concludes. Appendix A lists the Lean declarations; Appendix B gives the epistemic status table.

2 Historical Context: A Question Posed and Re-Posed

The discrete/continuum question has been asked in substantially different vocabularies by each generation that has taken it up, and the vocabulary shift is not cosmetic — each new formal tool changed what counted as an answer, and, we will argue, each generation’s tools eventually ran into some version of the same wall this paper formalizes.

2.1 The continuum default, and its first crack

Classical mechanics inherited its substrate assumption from geometry rather than from physics: space and time were modeled on the real continuum because that was the available mathematics, not because the continuum had been empirically established. Newtonian dynamics, and later field theory through general relativity, is written on differentiable manifolds with no built-in minimum length or minimum state count. For roughly three centuries, “the universe is continuous” was less a hypothesis than an unexamined modeling default, and the question this paper asks — can an internal observer decide it? — had no formal handle, because there was no rigorous notion yet of what a finite-resource decision procedure even was.

That handle appeared with computability theory in the 1930s (Turing, Church, Kleene), but its relevance to physics was not immediate. The connection was made explicit by Robin Gandy, who in 1980 asked, in effect, what physical principles a machine’s operation must satisfy for the Church–Turing thesis to extend from “mechanisms” in the abstract to devices built from actual matter [Gandy, 1980]. Gandy’s analysis — bounding the local density of information and the propagation speed of causal influence — was the first serious attempt to tie “how much information can be packed into a bounded physical region” (our axis A) to a computability claim about physical process (our axis B). It is a direct ancestor of both Bekenstein’s later entropy bound and of the present paper’s insistence that axes A and B, while conceptually distinct, are both instances of one underlying phenomenon: an internal observer’s access to the substrate is mediated entirely by a finite-alphabet readout channel, and every question about the substrate is really a question about what can be inferred from that channel.

2.2 The digital-physics tradition

Independently of Gandy, and starting earlier, a distinct tradition proposed that physical reality is not merely finite-information but literally computational in its substrate. Konrad Zuse’s 1969 *Rechnender Raum* (“Calculating Space”) [Zuse, 1969, 1970] is usually credited as the founding text: Zuse conjectured that the universe is being computed by a discrete, cellular-automaton-like process, decades before “digital physics” had a name and before Bekenstein or Bousso had given any information-theoretic content to “finite substrate information.” Edward Fredkin developed this into a research programme in the 1980s, proposing that reversible universal cellular automata are the correct microphysical model [Fredkin, 1990], and Stephen Wolfram’s *A New Kind of Science* [Wolfram, 2002] pushed the same idea into a large-scale programme of searching cellular automaton rule space for candidates that reproduce known physics. Gerard ’t Hooft took a related but distinct route from a different motivation: dissatisfied with the interpretive status of quantum mechanics, he proposed that quantum indeterminism is an emergent, coarse-grained description of an underlying deterministic, discrete (cellular-automaton-like) dynamics [’t Hooft, 1999, 2016] — so ’t Hooft’s discreteness claim is entangled with a specific stance on axis C (determinism) that Zuse, Fredkin, and Wolfram’s proposals do not require.

These positions share a structure worth naming precisely, because it is exactly the structure this paper shows cannot be internally vindicated: each proposes a *specific* discrete law or class of

discrete laws and treats successfully reproducing observed physics as evidence, or even proof, that the substrate literally is of that computational character. theorem 4 below says that no internal, finite-resource procedure can certify that a candidate law is the one generating the observed transcript forever, however well it has matched so far. This is not a criticism specific to the digital-physics tradition — the identical epistemic ceiling applies to any continuum theory claiming certification, including general relativity taken at face value as a claim about the actual substrate. It is a symmetric barrier, and one purpose of this paper’s abstraction (worlds, transcripts, tests, defined independent of any particular physical theory) is to make that symmetry a theorem rather than an assertion.

2.3 Complications from the computability side

The digital-physics tradition’s implicit dichotomy — discrete-and-computable versus continuum-and-mysterious — did not survive contact with two independent developments inside computability theory and analysis themselves, both of which this paper’s model deliberately inherits rather than glosses over.

First, David Deutsch’s 1985 formulation of the *physical* Church–Turing principle — that every finitely realizable physical system can be perfectly simulated by a universal computing machine operating by finite means [Deutsch, 1985] — was explicitly a physical claim, not a mathematical theorem, and Deutsch’s own paper already flagged the subtlety that a computable law can still support qualitatively new physical concepts (a quantum computer efficiently doing things intractable classically) without leaving the computable realm. The physical Church–Turing thesis being an empirically contentful claim about physical law, rather than a mathematical certainty, is precisely the kind of thing theorem 4 shows cannot be internally certified.

Second, and more corrosive to the naive dichotomy, Hava Siegelmann’s analysis of analog recurrent neural networks with real-valued (non-rational) weights showed that such networks can, in principle, compute functions no Turing machine can compute, evaluated over unbounded time [Siegelmann and Sontag, 1994, Siegelmann, 1999]. This is a genuine, mathematically rigorous hypercomputation candidate that lives entirely within continuum mathematics and requires no appeal to exotic new physics — it only requires that some physical quantity actually realize an uncomputable real number to unbounded precision. B. Jack Copeland’s philosophical taxonomy of hypercomputation [Copeland, 2002a,b] carefully separates several logically distinct routes to physical hypercomputation (accelerating Turing machines, infinite-precision analog computation, quantum hypercomputation proposals, oracle-consulting machines) that are frequently conflated in informal discussion; that taxonomy is one of the reasons this paper insists on stating precisely, and separately, what each of its two barriers does and does not assume about the observer’s computational resources (section 6, item 5).

Third, and most directly relevant to this paper’s modeling choice for the continuum class, Marian Pour-El and Ian Richards proved that the classical wave equation, with perfectly computable initial data, can have a unique solution that is *not* computable [Pour-El and Richards, 1981, 1989]. (A regularity caveat Penrose raised historically, and any careful referee will too: the non-computable solution arises from initial data that is computable but not C^1 ; with smooth, C^1 -computable data the solution *is* computable. This does not affect how the result is used below — definition 4.4 and section 10 only need the existence of *some* non-computable member of the computable-analysis-refined continuum class, which the non- C^1 construction still supplies.) This result is the single clearest formal demonstration that “continuum substrate” does not entail “hypercomputational substrate” and, symmetrically, that “discrete substrate” is not needed to obtain “computable substrate” — computability and cardinality of the state space are logically independent axes. It undercuts any

easy identification of continuum with dangerous-and-uncomputable or discrete with safe-and-tame, and it is exactly why this paper’s formal model (section 4) defines the discrete/continuum distinction by the cardinality and structure of the state space (`Substrate.DiscreteWorld`, `Substrate.ContinuumWorld`), not by whether the resulting transcript is computable, and why `Cstrict` — the class of continuum transcripts that happen to be non-computable — is defined as a strict subset of the continuum class rather than identified with it. A computable-analysis refinement of the continuum class (restricting to Cauchy-modulus-computable reals) would shrink `Cstrict`; by the Pour-El-Richards result it would not empty it, because their non-computable solution already arises from computable initial data under exactly that kind of restriction. We return to this in section 6, item 4.

2.4 The empirical side: how physics actually tried to test axis A

While the digital-physics tradition and the computability-theoretic complications developed largely in parallel, mainstream theoretical physics developed its own, independent answer to axis A — not by conjecturing a specific discrete mechanism, but by deriving an information bound from established physics. Jacob Bekenstein’s 1981 universal bound — that the entropy-to-energy ratio of any bounded physical system is bounded above by a quantity proportional to its radius, $S \leq 2\pi k_B ER/\hbar c$ — [Bekenstein, 1981] was derived from black-hole thermodynamics and the second law, not from any discreteness conjecture, and already gives a positive answer to axis A for systems of bounded energy: only finitely many distinguishable states fit in a bounded region of given energy content. Gerard ’t Hooft 1994 and Leonard Susskind 1995 then proposed the qualitatively stronger *area* bound — that the maximum entropy (equivalently, information content) of any bounded region is proportional to its *boundary area*, not its volume, at roughly one bit per Planck area — building explicitly on Bekenstein’s result but going beyond it: the area bound is a claim about geometry itself, not merely a bound tied to a system’s energy content [’t Hooft, 1994, Susskind, 1995]. Raphael Bousso generalized the area bound into a fully covariant form applicable to arbitrary spacetimes and causal structures [Bousso, 1999, 2002]. This is the historically important case of axis A being addressed *without* committing to any particular discretization scheme, computational interpretation, or even to the claim that spacetime itself is discrete — the holographic bound constrains information content, not geometry.

The natural next question — if information density is finite, does spacetime itself have to be a discrete lattice, and if so, wouldn’t such a lattice show up experimentally as a preferred frame (Lorentz violation)? — is where the causal-set programme becomes historically essential. Building on ideas that trace back through Bombelli, Lee, Meyer, and Sorkin’s 1987 proposal that spacetime is fundamentally a discrete partial order (a causal set) [Bombelli et al., 1987], Bombelli, Henson, and Sorkin proved in 2009 that a discrete causal set, generated by a Poisson sprinkling process, can be constructed to be exactly Lorentz invariant *in distribution* — no preferred rest frame, no direction picked out by the discretization [Bombelli et al., 2009]. This is a narrower and more precise claim than “no discreteness artifact whatsoever”: causal sets built this way still exhibit their own discreteness signatures (non-locality effects, so-called “swerves” in particle trajectories), which are themselves Lorentz-invariant rather than absent. The theorem rules out a preferred *frame*, not discreteness signatures altogether — a distinction worth stating precisely, because it makes this paper’s point sharper rather than weaker: it shows discreteness need not manifest as a symmetry violation at all, which is exactly why a naive “look for the grid” experiment is attacking a question of the wrong shape even before theorem 2’s general obstruction is invoked. This result mattered historically because it dissolved what had looked like a knock-down objection to any discrete spacetime theory (“a lattice necessarily picks out a preferred frame, we would have seen it by now”), and thereby made discrete-spacetime theories respectable candidates again.

We engage this history directly because it is the single clearest historical instance of exactly the phenomenon this paper’s Theorem 2 (theorem 2) proves in general. The naive Lorentz-violation test is precisely a finitely-determined property attempt: look for a specific, bounded-resource observable signature (anisotropic dispersion, a preferred direction) that a specific discretization scheme would produce. The causal-set programme’s own resolution — construct the discretization to be exactly Lorentz-invariant — is a demonstration, for one concrete scheme, that *that particular* finite test cannot distinguish the discrete case from the continuum case. theorem 2 generalizes this from “this one clever scheme evades this one clever test” to “no finite-resource test whatsoever, against any discretization scheme whatsoever, can settle the question, because the two transcript classes are dense into each other in Baire space.” The causal-set community’s Lorentz-invariance achievement is therefore best read, in retrospect, as a special-case patch that happens to defeat one attack; it was never, and could never have been on its own terms, a general resolution of the underlying undecidability, because no finite construction can be a general resolution of a statement about the density of two disjoint sets in an infinite product space.

2.5 Modern empirical programmes on axis B′, not axis B

Parallel to the causal-set programme’s engagement with axis A, two modern lines of work have tried to test whether physical dynamics can defeat known complexity-class bounds — axis B′ as introduced in section 1, not axis B itself, since neither line of work bears on whether dynamics is *computable* at all, only on how *efficiently* computable it is. Scott Aaronson’s “NP-complete problems and physical reality” [Aaronson, 2005] surveys and evaluates every serious proposal (from relativistic time dilation near a black hole to exotic quantum-gravitational effects) for physically solving NP-hard problems in polynomial time, concluding that none survives scrutiny and proposing, in effect, that nature’s apparent refusal to solve NP-hard problems efficiently is itself a physical principle worth taking as seriously as the second law of thermodynamics. Seth Lloyd’s estimates of the computational capacity of the observable universe [Lloyd, 2002, 2006] — bounding the number of elementary logical operations and bits the universe could have performed over its history, using the same Bekenstein-type bounds — are the natural quantitative complement: they do not test hypercomputation (axis B) at all, but put a number on how much classical computation “fits,” which is the natural backdrop against which any claimed axis-B′ anomaly (a physical process solving something outside BQP with polynomial resources) would have to be evaluated. Both of these are external-literature results about axis B′; the programme’s own internal result on axis B *itself* — not B′ — is NEMS Paper 35, “Oracles as External Selectors,” which proves a hypercomputation-exclusion taxonomy (no admissible internal procedure consults an oracle) directly from the Internality/Outsourcing Schema [Spivack, 2025m]; theorem 3 below complements it with a Lean-checked, model-theoretic certifier-silence result reached independently and by a different route.

Both Aaronson’s and Lloyd’s results are, in our terminology, attempts to test axis B′ empirically by looking for a single dramatic counterexample or bound violation; neither bears on axis B as this paper defines it, since a fast NP-hard solver is still a computable process. theorem 3 below (Theorem 3, sound certifier never fires) is nonetheless the right formal lens for how such a result should be interpreted even if it were found, on its own complexity-theoretic terms: by the same prefix-density argument (theorem 1) that proves theorem 3 for the discrete/continuum case, any finite observed prefix of “evidence for the violation” is equally consistent with an ordinary, bound-respecting computable continuation, so no internal test built from that finite prefix could be sound and ever fire on this kind of claim either — the formal reason a single demonstration is never internally certifiable, not the informal worry that it might be a resource-hiding artifact. Only the *scaling* of physical resource against instance size, across a family of instances, can be reported as evidence — never

certified as proof. This is why section 7 treats scaling exponents, not individual demonstrations, as the correct unit of empirical content for axis B', continuing the same methodological lesson the causal-set programme taught for axis A.

2.6 Where this paper sits

Every result surveyed above is either (i) a specific mechanism proposal (digital physics), (ii) a specific complication showing the discrete/continuum dichotomy does not cleanly track computability (Deutsch, Siegelmann, Pour-El-Richards), (iii) a specific information bound derived from established physics (Bekenstein, Bousso), or (iv) a specific empirical test or estimate (causal sets' Lorentz invariance, Aaronson's survey, Lloyd's estimates). This paper is none of these. It does not propose a mechanism, does not derive a new physical bound, and does not run or evaluate an empirical test. It proves, for an observer defined abstractly enough to cover every mechanism proposal and every empirical test above as a special case, that the entire *class* of such tests — not any one of them, all of them, uniformly — cannot settle the question. That is a claim about the logical structure of the problem, not a new entrant into the substantive debate about which substrate is correct, and it is the reason the paper is agnostic between Zuse/Fredkin/Wolfram/'t Hooft on one side and an unrestricted continuum view on the other: both sides are making a claim this paper proves can never be internally certified, regardless of which of them happens to be true.

3 The Three Axes and Prior Evidence

Axes A and B are exactly the two the theorems below show to be internally undecidable; axis C is genuinely different in kind and is addressed only by the glimmer corollary (section 8.1), whose definitions and separations are machine-checked but whose two remaining bridge premises are disclosed as open, not folded in as an established result. Axis B' (table 1) is also not addressed: it is a complexity-theoretic question about efficiency, not a computability question, and lies outside what theorem 4 or any other theorem below speaks to. The Bombelli–Henson–Sorkin causal-set result [Bombelli et al., 2009] is the standing witness that discreteness on axis A leaves no Lorentz-violating grid artifact, so the traditional Planck-scale experimental programme tests a particular scheme's symmetry properties, not substrate as such (section 2).

Remark 3.1 (Axis A is not literally D, and has its own dense pair). D as defined (definition 4.3) is an axis-B class — computability of transcript — not literally axis A's "finite states per bounded region." A finite-state world with a non-computable transition rule has finite information density (axis A) but a non-computable transcript, and a continuum world with a computable rule has the reverse; D tracks neither in isolation. To make the abstract's claim to cover "both principal readings" fully formal rather than only evidentially motivated (table 1, row A), axis A gets its own dense pair and its own instance of theorem 2: let $\mathbf{Dfin} := \{f \mid \exists w \text{ finite-state, } w.\text{transcriptFn} = f\}$ be the transcripts of worlds with finite state type. $\mathbf{Dfin}$ coincides exactly with the eventually periodic sequences ($\mathbf{Substrate.Dfin_iff_eventuallyPeriodic}$) — a pigeonhole argument on the finitely many reachable states forces eventual periodicity, and conversely every eventually periodic sequence is realized by a finite-state world built from its pre-period and period. $\mathbf{Dfin}$ is dense ($\mathbf{Substrate.Dfin_dense}$), its complement is dense ($\mathbf{Substrate.Dfin_compl_dense}$), and the two are disjoint by construction, so the abstract dense-pair form of Theorem 2 ($\mathbf{Substrate.not_finitely_determined_of_dense_pair}$, module $\mathbf{Substrate/NotFinDet.lean}$) applies verbatim to this different pair:

$$\neg \exists P, \text{FinitelyDetermined}(P) \wedge (\forall f \in \mathbf{Dfin}, P(f)) \wedge (\forall f \in \mathbf{Dfin}^c, \neg P(f))$$

Axis	Discrete pole	Continuum pole	Existing evidence	Attack surface
A. Information density	Finite states per bounded region	Infinitely many distinguishable states	Bekenstein’s energy bound + ’t Hooft/Susskind/Bousso’s holographic area bound $\Rightarrow$ finite [Bekenstein, 1981, ’t Hooft, 1994, Susskind, 1995, Bousso, 1999, 2002]	Exactness of the bound; metrological scaling
B. Dynamics	Computable (Church–Turing)	Non-computable / hypercomputation accessible	Siegelmann’s analog RNN construction shows hypercomputation is mathematically consistent [Siegelmann and Sontag, 1994, Siegelmann, 1999]; no confirmed physical instance	theorem 4: no internal law-certification, whether the candidate law is computable or not
B’. Efficient computability (<i>subsidiary to B — complexity, not computability; see section 1</i>)	P/BQP-computable with polynomial resources	NP-hard (or harder) solved with polynomial resources	Aaronson: nature refuses NP-hard in poly time [Aaronson, 2005]; Lloyd’s computational-capacity bounds [Lloyd, 2002, 2006]	Complexity class of physical computation — outside the Lean theorems below, which address computability, not efficiency
C. Chance	Deterministic	Objective randomness	Bell’s theorem, assuming no superdeterminism [Bell, 1964]; Hossenfelder–Palmer’s case for taking superdeterminism seriously is the standing challenge to that assumption, not support for it [Hossenfelder and Palmer, 2020]	Interacts with B via the “glimmer” programme (Ω -like traces at degree $\mathbf{0}'$); section 10

Table 1: The axes conflated by “is the universe discrete or continuous.” This paper’s Part I theorems (section 5) show axes A and B, *taken together* as the discrete-versus-strict-continuum distinction (section 4), are internally undecidable; definition 3.1 gives a separate, axis-A-specific dense pair ($\mathbf{Dfin}$, eventually periodic sequences) to which the same mechanism applies directly, machine-checked (definition 3.1). Axis B’ is a distinct, complexity-theoretic question (efficiency, not computability) that the theorems do not address and that is listed here only to correctly separate it from axis B’s existing evidence. Axis C is orthogonal and is addressed by the glimmer corollary (section 8.1): its definitions and separations are machine-checked, with two bridge premises (Born-martingale completeness and the degree- $\mathbf{0}'$ framework transfer) disclosed as open.

(`Substrate.axisA_not_finitely_determined`, module `Substrate/AxisA.lean`). Axis A is therefore internally undecidable in its own right, by the identical topological mechanism as axes A and B taken together, and not merely by inheritance from axis B’s D.

Lean status: `Substrate/AxisA.lean` is machine-checked — status `A_LEAN`. It is part of this repository’s green `Substrate.lean` import (`lake build`: 8262/8262 jobs, zero errors, zero `sorry`), and every theorem named above is confirmed by direct `#print axioms` to depend on at most the three standard Mathlib axioms (`propext`, `Classical.choice`, `Quot.sound`); see Appendix A for the full declaration table.

4 Formal Model

The formal development (Lean module `Substrate/Basic.lean`) fixes an abstraction of “physical system with an internal observer” deliberately thin enough to cover every substrate proposal in section 2 as a special case, and precise enough to support theorems.

4.1 Worlds and transcripts

Definition 4.1 (World). A *world* (`Substrate.World`) is a record consisting of a state type S , an initial state $\text{init} : S$, a deterministic step function $\text{step} : S \rightarrow S$, and a finite-alphabet observer readout $\text{read} : S \rightarrow \mathbb{N}$.

The readout is where “internal observer” enters formally, and it is the single most important modeling choice in the paper. Whatever the substrate actually is — finite state machine, continuum field configuration, or anything else — the observer receives only a stream of natural numbers, one symbol per time step. Everything an internal observer can possibly know about the substrate factors through this stream.

Definition 4.2 (Transcript). The *transcript* of a world w (`Substrate.World.transcript`, `Substrate.World.transcriptFn`) is the function $\mathbb{N} \rightarrow \mathbb{N}$ given by

$$w.\text{transcript}(t) = w.\text{read}(w.\text{step}^{[t]}(w.\text{init})).$$

The transcript is the *only* thing the observer ever sees; every downstream statement in this paper is a statement about elements of Baire space $\mathbb{N} \rightarrow \mathbb{N}$.

4.2 Two substrate classes, defined by what they can produce

Substrate classes are defined by the transcripts they can produce, not by the state type directly — this is what makes the indistinguishability theorems (section 5) clean, and it is a deliberate anti-smuggling choice: a definition of “discrete” phrased in terms of the state type alone would risk building the conclusion into the premise.

Definition 4.3 (Discrete and continuum worlds). A world w is *discrete* (`Substrate.DiscreteWorld`) if its state type admits a `Primcodable` structure and both `step` and `read` are `Computable`. A world w is a *continuum* world (`Substrate.ContinuumWorld`) if its state type is isomorphic to $\text{Fin } n \rightarrow \mathbb{R}$ for some n , with *arbitrary* (unrestricted classical) dynamics.

$$\mathbf{D} := \{f \mid \exists w \text{ discrete, } w.\text{transcriptFn} = f\} \quad \mathbf{C} := \{f \mid \exists w \text{ continuum, } w.\text{transcriptFn} = f\}$$

$$\mathbf{Cstrict} := \mathbf{C} \setminus \{f \mid \text{Computable } f\}$$

(`Substrate.D`, `Substrate.C`, `Substrate.Cstrict`.)

Remark 4.4 (Modeling honesty). `ContinuumWorld` places no computability restriction whatsoever on step; consequently `C` turns out to be *all* of Baire space (`Substrate.mem_C`, section 5): an unrestricted continuum with arbitrary classical dynamics can realize any readout stream. This is a deliberate and disclosed modeling choice, not an oversight — it reflects the historical fact (section 2) that “continuum” as classically conceived carries no built-in computability restriction, and it is exactly why the interesting content of the continuum class lives in `Cstrict`, the sub-class that is *also* non-computable. A computable-analysis refinement of `ContinuumWorld` — restricting step to act by Cauchy-modulus-computable maps on Cauchy-modulus-computable reals — would shrink `Cstrict`, but by Pour-El–Richards [Pour-El and Richards, 1981] would not empty it, since their non-computable wave-equation solution already arises from computable initial data under exactly that kind of restriction. Mathlib does not currently provide computable analysis, so this refinement is left for future work (section 10); the theorems below do not depend on it.

4.3 Internal tests and finite determination

Definition 4.5 (Prefix, internal test). For $f : \mathbb{N} \rightarrow \mathbb{N}$ and $n \in \mathbb{N}$, `Prefix(f, n)` (`Substrate.Prefix`) is the list of the first n values of f . An *internal test* (`Substrate.InternalTest`) is a pair of a budget $n \in \mathbb{N}$ and a computable verdict function `verdict : List ℕ → Bool`; it runs (`Substrate.InternalTest.run`) as $T.\text{run}(f) := T.\text{verdict}(\text{Prefix}(f, T.\text{budget}))$.

Definition 4.6 (Finitely determined). A property P of transcripts is *finitely determined* (`Substrate.FinitelyDetermined`) if some prefix length n settles it for every pair of transcripts agreeing on that prefix:

$$\exists n, \forall f g, \text{Prefix}(f, n) = \text{Prefix}(g, n) \Rightarrow (P(f) \leftrightarrow P(g)).$$

Equivalently, P is clopen in the product topology on $\mathbb{N} \rightarrow \mathbb{N}$.

An internal test decides a property exactly when the property is internally decidable (section 8, `Substrate.InternallyDecidable`); every internally decidable property is finitely determined (`Substrate.internally_decidable_implies_finitely_determined`), since a test only ever inspects a bounded-length prefix.

5 Theorems, Part I: The Two Barriers

All results in this section are proved and machine-checked (status `A_LEAN`) in the Lean module family `Substrate/`, against Mathlib only.

Lemma 2.1 (Transcript characterisation of D). $f \in D \iff \text{Computable } f$. (`Substrate.transcript_characterisation`, module `Substrate/Discrete.lean`.)

Proof sketch. ($\Rightarrow$) If w is discrete, iterating a computable step from a computable state is computable (`Substrate.World.iterate_computable`, via `Computable.nat_rec`), and composing with computable read shows the transcript is computable (`Substrate.World.transcript_computable_of_discrete`). ($\Leftarrow$) Given computable f , take $S := \mathbb{N}$, $\text{step} := \text{succ}$, $\text{init} := 0$, $\text{read} := f$; this world is discrete by construction and its transcript is f by $\text{succ}^{[t]}(0) = t$. $\square$

Lemma 2.2 (Embedding). *Every discrete world’s transcript is realized by some continuum world; hence $D \subseteq C$.* (`Substrate.discrete_subset_continuum`, module `Substrate/Embed.lean`.)

Proof sketch. This is the immediate specialisation of the stronger fact, proved directly, that *every* $f : \mathbb{N} \rightarrow \mathbb{N}$ whatsoever lies in $\mathbf{C}$ (`Substrate.mem_C`): since $\mathbb{N} \rightarrow \mathbb{N}$ and $\mathbb{R}$ have the same cardinality (`Cardinal.mk` agreement, via $\aleph_0^{\aleph_0} = \mathfrak{c}$), `Mathlib's Cardinal.eq` converts that cardinality equality into a *classical*, non-constructive equivalence $(\mathbb{N} \rightarrow \mathbb{N}) \simeq (\text{Fin } 1 \rightarrow \mathbb{R})$ — this step is where `Classical.choice` (and, via `Cardinal.mk's` quotient construction, `Quot.sound`) does genuine work in this proof, not merely disclosed bookkeeping: no *explicit*, computable equivalence between the two types is exhibited or exhibitable by this argument. Given that equivalence, take the continuum world's state type to be $\mathbb{N} \rightarrow \mathbb{N}$ itself, `init` := f , step the left-shift, and read the head symbol; its transcript is exactly f . $\square$

Theorem 1 (Prefix Indistinguishability). *For every finite list p , there exist $d \in \mathbf{D}$ and $c \in \mathbf{Cstrict}$, both extending p as a prefix: d is p followed by zeros (computable), and c is p followed by a fixed non-computable tail built from the halting problem.* (`Substrate.prefix_indistinguishable`, module `Substrate/Prefix.lean`.)

Proof sketch. The discrete witness is the zero-extension of p , computable by construction (`Substrate.prefix_getD_eq`). The strict-continuum witness agrees with p on the prefix and then continues with a fixed function `haltingIndicator` : $\mathbb{N} \rightarrow \mathbb{N}$ (`Substrate.haltingIndicator`), defined as the (classically well-defined, non-effective) indicator of whether a code halts on input 0; this function is provably not computable (`Substrate.not_computable_haltingIndicator`), by reduction from `ComputablePred.halting_problem`, and this non-computability transfers to the prefix-extended witness. By lemma 2.2 the witness also lies in $\mathbf{C}$, hence in $\mathbf{Cstrict}$. $\square$

Corollary 1.1. *$\mathbf{D}$ and $\mathbf{Cstrict}$ are disjoint (`Substrate.D_disjoint_Cstrict`, an immediate consequence of lemma 2.1) and both dense in Baire space (immediate from theorem 1, since every finite prefix extends into both classes).*

Theorem 2 (Substrate is not finitely determined). *No finitely determined property of transcripts holds on all of $\mathbf{D}$ and fails on all of $\mathbf{Cstrict}$:*

$$\neg \exists P, \text{FinitelyDetermined}(P) \wedge (\forall f \in \mathbf{D}, P(f)) \wedge (\forall f \in \mathbf{Cstrict}, \neg P(f)).$$

(`Substrate.substrate_not_finitely_determined`, module `Substrate/NotFinDet.lean`.)

Proof sketch. Immediate from theorem 1: if P were settled by prefix length n , apply theorem 1 to the all-zero prefix of length n to get $d \in \mathbf{D}$ and $c \in \mathbf{Cstrict}$ agreeing on that prefix; then $P(d) \leftrightarrow P(c)$ contradicts $P(d)$ true and $P(c)$ false. $\square$

This is the headline result — and it does not need the halting problem

Theorem 2 is pure topological density in Baire space. Its proof invokes theorem 1, whose **Cstrict** witness happens to be built from a halting-problem construction, but nothing about Theorem 2 *itself* depends on recursion theory: any two dense, disjoint subsets of $\mathbb{N} \rightarrow \mathbb{N}$ would give the same conclusion. This is a Lean-checked fact, not only a proof-sketch observation: the abstract form

$$(\forall p, \exists a \in A, \text{Prefix}(a, |p|) = p) \rightarrow (\forall p, \exists b \in B, \text{Prefix}(b, |p|) = p) \rightarrow \text{Disjoint } A \ B \rightarrow \\ \neg \exists P, \text{FinitelyDetermined}(P) \wedge (\forall f \in A, P(f)) \wedge (\forall f \in B, \neg P(f))$$

is proved directly from prefix combinatorics, with no reference to **Computable** or **Nat.Partrec** anywhere in its proof term (`Substrate.not_finitely_determined_of_dense_pair`, module `Substrate/NotFinDet.lean`; `#print axioms` shows only the standard `propext`, `Classical.choice`, `Quot.sound`), and theorem 2 itself is derived from it by instantiating $A := D$, $B := \text{Cstrict}$ via theorem 1 (`Substrate.substrate_not_finitely_determined`). The barrier is topological *before* it is recursion-theoretic, and Lean shows this at the level of the proof term, not only the theorem statement. This separation is developed further in section 6, item 5, and is one of the two independent barriers advertised in the abstract.

Theorem 3 (No sound internal certifier fires). *Let T be an internal test.*

- (a) *If T is sound for continuum (`Substrate.SoundForContinuum`: $T.\text{run}(f) = \text{true} \Rightarrow f \in \text{Cstrict}$), then T never returns true (`Substrate.sound_for_continuum_never_fires`).*
- (b) *Symmetrically, if T is sound for discreteness (`Substrate.SoundForDiscreteness`), T never returns true (`Substrate.sound_for_discreteness_never_fires`).*

(Combined: `Substrate.certifier_silent_or_unsound`, module `Substrate/Certifier.lean`.)

Proof sketch. Both directions apply theorem 1 to the prefix that T actually inspects ($\text{Prefix}(f_0, T.\text{budget})$ for a hypothetical positive verdict f_0): the **D**-witness and the **Cstrict**-witness agreeing with that prefix both trigger the same verdict, so soundness for one class forces the (false) claim that the other class’s witness also lies in the sound class, contradicting corollary 1.1. $\square$

A certifier of either substrate class is therefore either unsound (fires on the wrong class at least once) or permanently silent (never fires at all) — there is no third option. This is the model-theoretic counterpart of NEMS Paper 35’s programme-internal “no internal hypercomputer” result [Spivack, 2025m] (section 2): Paper 35 argues from the Internality/Outsourcing Schema that no admissible internal procedure may consult an oracle, an existence claim about hypercomputers; theorem 3 complements it with a distinct, certifier-silence conclusion for this paper’s specific model, by the independent route of prefix-density in Baire space.

Theorem 4 (Law-Certification Barrier). *Let f be computable, and let $\text{LawMatches}(f, e)$ (`Substrate.LawMatches`) say that code e matches f at every time step: $\forall t, e.\text{eval}(t) = \text{Part.some}(f(t))$. Then $\text{LawMatches}(f, \cdot)$ is not a computable predicate on codes:*

$$\neg \text{ComputablePred}(\lambda e. \text{LawMatches}(f, e)).$$

(`Substrate.law_match_undecidable`, module `Substrate/LawBarrier.lean`.)

Proof sketch. By reduction from the halting problem, reusing the halting-reduction *pattern* of `nems-lean`’s `NemS.Diagonal.HaltingReduction` [Spivack, 2025r] (no code dependency; Mathlib only). Given a code c , build a code e_c (`Substrate.matchFn`, `Substrate.bigG`) that, on input t , outputs $f(t)$ unless c has *visibly* halted on input 0 within t steps (checked via the step-bounded evaluator `evaln`), in which case it outputs $f(t)+1$ (`Substrate.matchFn_eq_f_iff`). Then e_c matches f forever if and only if c never halts on 0. The assignment $c \mapsto e_c$ is computable, uniformly, via the S_n^m theorem (`Nat.Partrec.Code.curry`). If `LawMatches`($f, \cdot$) were a computable predicate on codes, composing with $c \mapsto e_c$ would computably decide $\neg \text{HaltsOnZero}(c)$ for every c , hence would decide the halting problem, contradiction. $\square$

Corollary 4.1 (Bridge to Paper 30). *Specializing theorem 4 to the constant transcript $f \equiv 0$ gives $\neg \text{ComputablePred}(\lambda e. \text{LawMatches}(0, e))$ (`Substrate.law_match_undecidable_zero`). This is the substrate-specific instance of the general no-total-self-certifier phenomenon proved by NEMS Paper 30, “Second Incompleteness for Self-Certifying Learners” [Spivack, 2025k]: no total internal certifier exists under diagonal capability. Substrate law-certification is thus a special case of self-certification failure, not a new phenomenon.*

Interpretation: two closures, not one

Theorem 2 says no finite *data* decides substrate. Theorem 4 says no finite *theory* can be internally certified either, even when the theory is handed to the observer outright. Together they close both the evidential route and the law-based route; the second closure is the diagonal barrier proper, and it is the one that genuinely needs computability theory.

Theorem 4’ (Π_2^0 -hardness of `LawMatches`). *Let f be computable. The totality problem $\text{Tot} := \{e \mid \forall t, e.\text{eval}(t) \text{ halts}\}$ many-one reduces to `LawMatches`($f, \cdot$):*

$$\text{Tot} \leq_m (\lambda e. \text{LawMatches}(f, e)).$$

Since Tot is Π_2^0 -complete — a standard recursion-theoretic fact, cited here and not proved in this repository — this pins `LawMatches`($f, \cdot$)’s complexity exactly at Π_2^0 -complete, strictly above the co-halting lower bound Theorem 4 gives alone. (`Substrate.law_match_pi2_hard`, specialized to $f \equiv 0$ as `Substrate.law_match_pi2_hard_zero`; module `Substrate/LawBarrier.lean`.)

Proof sketch. Given a code e , build a code e' that, on input t , first runs e on that same input t to completion (discarding the result) and then outputs $f(t)$; this sequential-composition construction is computable, uniformly in e . If $e \in \text{Tot}$ (e halts on every input), e' reaches its final step on every t and outputs $f(t)$ there, so e' matches f forever; if $e \notin \text{Tot}$, e fails to halt on some particular input t_0 , so e' never reaches its final step on t_0 and fails to match f there. This gives $\text{Tot} \leq_m \text{LawMatches}(f, \cdot)$ directly, independently of Theorem 4’s own co-halting reduction. $\square$

6 What the Theorems Do Not Say

This section exists because the results above are easy to overstate, and an easy target for a referee who reads past the theorem statements. We state the limits precisely.

1. **The theorems do not say the universe is discrete.** They say the question cannot be settled internally, by evidence (theorem 2) or by law-certification (theorem 4), and they identify what *can* adjudicate instead (section 8).

2. **The theorems do not fully decide axis C (randomness).** A discrete deterministic world and a continuum world with objective chance can produce statistically identical Bell correlations; nothing above distinguishes them. Axis C is addressed directly by the glimmer corollary (section 8.1): its definitions, its density/separation-from-D topology, and its own instance of theorem 2 are machine-checked in a companion Lean module (`Substrate/Glimmer.lean`, part of the audited, sorry-free build); the finite-description consequence of theorem 5 depends on this class existing at all, so section 8.1 states it as settled content rather than an open problem. What remains genuinely open, independently of that Lean status, are **two** distinct premises. The **Born-martingale lemma**: that the actual physical Born measure, as complete internal semantics per NEMS Papers 13/14 [Spivack, 2025e,f], admits no successful computable martingale on the realized transcript, given the non-degeneracy premise of NEMS Paper 12 [Spivack, 2025d]. The **framework-transfer premise**: that the degree bound of UGP Paper 51 [Spivack, 2026] — proved there for the GTE adjudication layer under that paper’s own named premise bundle, not for an arbitrary framework’s transcripts — transfers to this paper’s framework-agnostic World model. We state the dependency chain honestly in section 8.1 and do not claim more of the corollary than is actually established: the mathematical machinery (the definitions, the separations, and the elementary bet-everything martingale fact) is machine-checked; the two physical/interpretive premises connecting it to the actual Born measure and to UGP Paper 51’s degree bound are not.
3. **The theorems do not say discreteness is unfalsifiable in every sense.** theorem 2 rules out *finitely determined* (clopen) properties, but asymptotic properties — limits over unboundedly long observation, not settled by any fixed prefix length — are not excluded and are in principle limit-testable (section 7). The theorem locates the only place substrate information can live; it does not eliminate all empirical content.
4. **The continuum class in this v1 development is classical and unrestricted**, so `C` is all of Baire space (definition 4.4). A computable-analysis refinement would shrink `Cstrict`, not empty it, by Pour-El–Richards [Pour-El and Richards, 1981, 1989]: their non-computable wave-equation solution already arises from computable initial data, so even a maximally restrictive computable-analysis reading of “continuum” still admits non-computable transcripts. The results are robust to this refinement; we have not carried it out because Mathlib does not yet provide computable analysis (section 10).
5. **Theorem 2 does not depend on Church–Turing; Theorem 4 does, and the oracle strength needed to escape it is pinned down exactly.** theorem 2 is pure density in Baire space (the boxed remark after its proof) and would hold verbatim for any two dense, disjoint transcript classes, computable or not; no oracle, at any level, changes which prefixes exist in `D` and `Cstrict` or the density fact between them, so theorem 2 is not escaped by *any* amount of extra computational power. theorem 4 is a genuine diagonal argument that requires the observer’s internal test to be a computable predicate — and the oracle strength that defeats it is exactly $\mathbf{0}''$, not $\mathbf{0}'$. $\text{LawMatches}(f, \cdot)$ is a Π_2^0 predicate on codes (a universal claim, over every time step, that evaluation halts *and* agrees with f); the reduction proving theorem 4 gives co-halting-hardness, and a second reduction (`Substrate.law_match_pi2_hard`: the totality problem $\text{Tot} := \{e \mid \forall t, e.\text{eval}(t) \text{ halts}\}$ many-one reduces to $\text{LawMatches}(f, \cdot)$, via a code that runs the candidate on each input to completion and discards the result) establishes Π_2^0 -hardness — strictly above the co-halting lower bound alone, and, since Tot is Π_2^0 -complete (a standard, cited recursion-theoretic fact, not proved in this repository), this pins $\text{LawMatches}(f, \cdot)$ ’s complexity exactly at Π_2^0 -complete. A $\mathbf{0}''$ oracle (halting relative to $\mathbf{0}'$, sufficient to decide any Π_2^0 statement)

escapes theorem 4; a plain halting ($\mathbf{0}'$) oracle provably does *not*, since no Π_2^0 -complete problem is decidable relative to $\mathbf{0}'$ — superseding the earlier, weaker claim that a $\mathbf{0}'$ oracle was merely “not shown” to escape. What *is* established, and is worth stating explicitly, is that theorem 2 survives strictly more observer upgrades than theorem 4: no oracle escapes the former, while exactly $\mathbf{0}''$ -and-above escapes the latter.

7 Where Substrate Information Can Live: Scaling Exponents

theorem 2 rules out finitely determined (clopen) observables as substrate evidence. It does not rule out *asymptotic* observables — quantities defined as a limit over an unboundedly growing resource budget, which by construction are not settled by any fixed prefix length. This section classifies the known candidates (status: discussion-level, “D” in Appendix B, not machine-checked) and explains, for each, why it is the correct *kind* of empirical target under the barriers proved above.

Metrological scaling. In an unrestricted continuum, achievable measurement precision scales without a floor as the number of resources (e.g. repeated trials or entangled probes) N grows — the Heisenberg $1/N$ scaling law. Under finite substrate information, this scaling exponent must break at some resource level, because there are only finitely many distinguishable outcomes to resolve into. The signature is an exponent measured over decades of resource scale, not a sharp threshold at any single N — which is exactly the form theorem 2 allows. This signature is forced by discreteness of *any* kind, unlike a Lorentz-violation search, which tests only one specific discretization scheme (section 2).

Spectral type and recurrence. A finite-dimensional Hilbert space forces pure point spectrum, exact quasi-periodicity, and a finite Poincaré recurrence time; a continuum Hilbert space permits continuous spectra and, generically, no finite recurrence time. This is observable, in principle, as a recurrence-time scaling law with system size, again an asymptotic rather than a threshold signature.

Complexity class of physical computation (axis $\mathbf{B}'$, not axis $\mathbf{B}$). Any physical process solving a problem outside BQP with polynomial physical resources would be evidence on axis $\mathbf{B}'$ — efficient computability (section 1, table 1) — not axis $\mathbf{B}$: such a process remains computable in the ordinary sense, so it says nothing about membership in **Cstrict** and does not itself bear on theorem 4. It is nonetheless the natural empirical target for Aaronson’s programme [Aaronson, 2005] (section 2). By the same prefix-density argument (theorem 1) underlying theorem 3: a *single* such demonstration can never be internally certified as a genuine complexity-class violation, since any finite observed prefix of the demonstration is equally consistent with an ordinary, bound-respecting computable continuation, so no internal test built from that prefix could be sound and ever fire. Only the *scaling* of resource against instance size, across many instances, can be reported as evidence — never certified as proof.

Constants. Finite decimal expansions of physical constants decide nothing: theorem 2 says any finite data is compatible with both D and **Cstrict**. A theory that predicts a constant from a finite combinatorial structure and keeps being empirically confirmed to increasing precision is MDL *evidence* in the sense of theorem 5 — never certification in the sense of theorem 4. This is explicitly the correct epistemic status to claim for predictions coming out of the Universal Generative Principle (UGP) programme, and we state it plainly: such predictions, however striking, are evidence of the

kind Theorem 5 identifies as the unique admissible kind, not proof of the kind Theorem 4 shows to be structurally unavailable.

8 Theorems, Part II: Outsourcing and the Unique Admissible Adjudicator

Definition 8.1 (Internally decidable). A transcript property P is *internally decidable* (`Substrate.InternallyDecidable`) if some internal test decides it exactly: $\exists T, \forall f, T.\text{run}(f) = \text{true} \leftrightarrow P(f)$.

Theorem 5 (Outsourcing and the Unique Adjudicator). (a) ***Outsourcing*** (*Lean-checkable half*). No internally decidable property separates D from $C\text{strict}$:

$$\neg \exists P, \text{InternallyDecidable}(P) \wedge (\forall f \in D, P(f)) \wedge (\forall f \in C\text{strict}, \neg P(f)),$$

and symmetrically for the reverse separation (`Substrate.substrate_not_internally_decidable`, `Substrate.substrate_not_internally_decidable'`). Consequently, neither $f \in D$ nor $f \in C\text{strict}$ is itself internally decidable (`Substrate.discrete_not_internally_decidable`, `Substrate.strict_continuum_not_internally_decidable`). (Module `Substrate/Outsourced.lean`; status `A_LEAN`.)

(b) ***Uniqueness of the MDL adjudicator (cited, not re-derived)***. The only adjudication procedure consistent with *PSC* — one that (i) consults no oracle and (ii) is invariant under presentation of the same law-set — is selection over complete law-sets by minimum description length (*MDL*). (Argument cited to NEMS Paper 34, “A Sieve Engine for Theory Spaces,” and Paper 84, “The Survivor Calculus” [Spivack, 2025l,q]; status `A_MDL`, not `A_LEAN` — this half is not a Mathlib reduction and is not re-derived here.)

Proof sketch, part (a). Every internally decidable property is finitely determined (`Substrate.internally_decidable_implies_finitely_determined`: a test only ever inspects a bounded prefix), so part (a) follows immediately from theorem 2. Part (b) is cited to the NEMS programme results named above, not re-derived in this repository. $\square$

Under Perfect Self-Containment (*PSC*), a self-contained system has no external selector to consult: there is no oracle, and no observer outside the system, that could supply a verdict the system itself cannot compute. theorems 2 and 4 together show that substrate class is not internally decidable from evidence or from law-certification — part (a) above. Hence substrate discreteness is an *outsourced* property in the precise sense of the Internality/Outsourcing Schema, NEMS Paper 83 [Spivack, 2025p]: a load-bearing task (here, adjudicating substrate) that cannot be discharged by any procedure internal to the system doing the adjudicating — which is exactly what makes part (b)’s external adjudicator necessary.

Consequence: D consists exactly of the transcripts admitting a finite description (by lemma 2.1, a computable one); $C\text{strict}$ by construction admits none. MDL therefore adjudicates, uniquely and non-arbitrarily, in favor of the **finite-description class** ($D \cup \text{Glimmer}$), not D alone — not as a methodological preference for simplicity, but as the one selection procedure that survives the closure requirement *PSC* imposes. D alone would be the wrong name for what MDL selects: NEMS Paper 12 (Determinism No-Go) proves that no total computable law generates the records in a diagonal-capable *PSC* universe with genuine record-divergent choice [Spivack, 2025d], so a universe with that property has no computable transcript and hence is not itself a member of D ; the glimmer class (Δ_2^0 , finitely described by law plus one jump, section 8.1) is what closes this gap, and is why

section 8.1 is a load-bearing part of this theorem’s statement rather than a deferred open problem. This is the bridge to the Universal Generative Principle (UGP) programme: UGP’s stance that a compact, finite generative structure functions as a certificate for physical law is precisely theorem 5 applied to one specific law-set, and section 7’s remark on constants states the resulting epistemic status explicitly.

“Discrete” is a description-length notion, not a geometry, and a continuum substrate can be its worked example. One clarification matters for a physicist reader applying theorem 5 to a candidate fundamental theory: *“discrete” in this paper never meant “lattice”; it meant “finitely generated.”* D is defined by computable *transcript* (lemma 2.1), a description-length class, not a geometry class, and lemma 2.2 already proves $D \subseteq C$: every discrete-transcript world is realized by *some* continuum world. A continuum substrate with a finite arithmetic generator therefore sits in $D \cap C$ by transcript — continuum by state type, finitely described by transcript — and this is not a counterexample to theorem 5; it is the theorem’s worked example. The Universal Generative Principle (UGP) programme’s Φ_{MDL} /GTE framework is exactly this case: a continuum field whose observable transcript is generated by a finite arithmetic certificate, so it is precisely the kind of world theorem 5 selects, stated in the vocabulary of a specific physical theory rather than in the abstract model above — not a rival to this paper’s agnosticism, but an instance of what the theorem already predicts should be favored.

This also resolves an apparent tension with NEMS Paper 24’s claim that the universe must begin as a discrete, self-interpreting “Reflexive Seed” [Spivack, 2025i]. That claim is a theorem 5-direction result about *initial conditions*, not an internal certification of the theorem 4 kind: a classical singularity is an underspecified boundary that requires external completion, hence non-foundational under PSC, so closure prefers a finite, self-interpreting origin over an unbounded one [Spivack, 2025g,j]. Crucially, the seed is prior in *generative* order, not *temporal* order. The Infinity Compression thesis’s two-layer structure — a bare certificate B , an enriched realization E , and a comparison map $\pi : E \rightarrow B$ with nontrivial fibers and sections — is the formal handle for exactly this relation: the finite generator is B , the continuum realization is E , and asking whether B exists “before” E is asking whether a genotype exists before its phenotype (an analogy offered with its own caveat, since it is not exact: a genotype is itself a physical object in time, whereas the seed is a type) — generative priority, not temporal priority. On this reading, Paper 24’s discreteness claim and this paper’s substrate-agnosticism are consistent rather than in tension: Paper 24 makes a claim about admissible initial data (an outsourcing/MDL selection, in this paper’s theorem 5 sense), not a claim that spacetime itself is a lattice, and not an internal certification that would run afoul of theorem 4.

Corollary 5.1 (Uncountable-observer strengthening of Theorem 5). *D is countable (`Substrate.D_countable`) and C_{strict} is not (`Substrate.Cstrict_not_countable`). More strongly, no family of internal tests $tests : I \rightarrow \text{InternalTest}$, indexed by an arbitrary type I — including an uncountable one — jointly decides discrete membership:*

$$\neg \exists (tests : I \rightarrow \text{InternalTest}), (\forall f \in D, \exists i, tests(i).run(f)) \wedge (\forall f \in C_{strict}, \forall i, \neg tests(i).run(f)).$$

(`Substrate.no_joint_substrate_decision`, module `Substrate/Cardinality.lean`; an analogous strengthening for internally decidable predicates, over any index type, is `Substrate.no_joint_internally_decidable`.)

Proof sketch. The obstruction is exactly theorem 1 again, not a cardinality argument: fix any candidate family and let f_0 be the constant-zero (discrete) transcript; some test i must accept f_0 . Applying theorem 1 to the prefix that test i actually inspects produces a **Cstrict**-witness agreeing with f_0 on that prefix, so test i accepts it too, contradicting the joint-decision hypothesis. No step of this argument depends on the cardinality of the index type I ; an uncountable family of observers adds no power beyond a single one (`Substrate.uncountable_index_offers_no_power`). The countability facts above are worth recording in their own right — D is a subset of the computable functions, hence countable (`Substrate.computable_set_countable`), while **Cstrict** occupies almost all of Baire space (`Substrate.baire_space_not_countable`) — but they are not what does the work here; the work is prefix indistinguishability, exactly as in theorems 2 and 5. $\square$

This forecloses a natural escape attempt one might otherwise propose against theorem 5: pooling an unlimited, even uncountable, supply of cooperating internal observers cannot recover what a single observer cannot decide, since the obstruction is topological density in Baire space, which is entirely insensitive to how many observers are asking.

Corollary 5.2 (Measure-theoretic typicality of the continuum class). *Under the canonical geometric i.i.d. product measure μ_p on Baire space (any parameter $p \in (0, 1)$), D has measure zero and **Cstrict** has full measure:*

$$\mu_p(D) = 0, \quad \mu_p(\text{Cstrict}) = 1.$$

(`Substrate.D_null_geometric`, `Substrate.Cstrict_full_mass_geometric`, module `Substrate/Probabilistic.lean`.)

Proof sketch. μ_p has no atoms: the per-coordinate geometric measure of any single symbol is bounded above by $p < 1$ uniformly, so the infinite product of these bounds tends to 0, giving $\mu_p(\{f\}) = 0$ for every transcript f . Since D is countable (corollary 5.1), it is a countable union of μ_p -null singletons, hence itself μ_p -null (`Substrate.D_null_geometric`). Since $C = \text{univ}$ (definition 4.4, `Substrate.C_eq_univ`), `Cstrict` = `univ` \ D (`Substrate.Cstrict_eq_univ_diff_D`), so $\mu_p(\text{Cstrict}) = 1 - \mu_p(D) = 1$ (`Substrate.Cstrict_full_mass_geometric`). $\square$

This reinforces corollary 5.1 from a different direction: not only does no family of internal observers, of any cardinality, jointly decide substrate class, but under the geometric family of product measures specifically, a transcript drawn at random is strict-continuum with probability 1. That verdict is not, however, forced by the mathematics of measure itself, and stating it as typicality *simpliciter* would be exactly the external-selector move section 1.1 identifies as PSC’s target: choosing μ_p over every other measure on Baire space is itself an unexplained external choice, on a par with eternal inflation’s measure problem. The clearest way to see this is to change the choice. The Solomonoff/MDL universal prior — section 1.1’s own example of PSC applied to constants, and the actual adjudication principle theorem 5 invokes — weights a sequence’s finite prefixes by description length rather than by a fixed per-symbol probability: a computable sequence generated by a program of size c has prefix-length- n algorithmic probability bounded below by $2^{-c-O(\log n)}$, a decay that is polynomial in n , while *every* sequence’s length- n prefix, computable or not, decays exponentially in n under any nondegenerate μ_p — the identical atomlessness mechanism behind this corollary’s own proof [Li and Vitányi, 2008]. The universal prior itself depends on a choice of reference machine, but only up to an additive constant fixed by the pair of machines being compared, not by the sequence under evaluation — the invariance theorem’s own guarantee [Li and Vitányi, 2008] — which is exactly presentation-invariance, theorem 5’s condition (ii), applied to the choice of universal prior itself rather than to a choice among law-sets: the universal prior is therefore an admissible ingredient for the kind of closure-respecting adjudication theorem 5 already requires, not

a second, unexamined external choice smuggled in alongside μ_p 's. Under the universal prior, then, finitely describable sequences are the ones exponentially favored, not the ones typicality discards: corollary 5.2's verdict and the universal prior's verdict are exact opposites, and the entire disagreement between them lives in which prior the adjudicator adopts, not in any fact about the substrate. This is the sharpest illustration available of why theorem 5 adjudicates the way it does: it does not sample from μ_p , from the universal prior, or from any measure over Baire space at all, precisely because doing so would smuggle back in the external selector PSC forbids. MDL selection over complete law-sets is a closure-respecting procedure, not a draw from a distribution over which universe is real, and corollary 5.2 is best read not as evidence for the continuum but as a demonstration of exactly how badly a naive sampling-measure adjudicator fails the very closure requirement this paper's actual adjudicator is built to satisfy. None of this bears on what an *internal test* can detect from a finite prefix, nor does it by itself yield an ε -tolerant test: see the probabilistic-observers discussion in section 10 for exactly where that further step remains open and why.

8.1 The glimmer corollary

The glimmer corollary is essential to Theorem 5, not an optional addendum to it: without it, the theorem's conclusion would name a class that the programme's own account of axis C says the actual universe is not in. NEMS Paper 12 (Determinism No-Go) proves that no total computable law generates the records in a diagonal-capable PSC universe with genuine record-divergent choice [Spivack, 2025d], so a universe with that property does *not* have a computable transcript, and D alone would then be the wrong name for what MDL actually selects. Glimmer resolves this: a glimmer world has a *computable law* and a Δ_2^0 transcript — the outcomes are not free bits, since they are fixed by the law plus one jump with no external selector, so the world still has a finite description and theorem 5 still selects it. section 8's consequence paragraph and the abstract accordingly say “finite-description class ($D \cup \text{Glimmer}$)” rather than “finite-information class,” for exactly this reason. The definitions and theorem statements below are proved in a companion Lean module, `Substrate/Glimmer.lean`, **machine-checked and part of this repository's audited, sorry-free build** (full Lean status disclosed below, after the derivation chain); the classification itself is a matter of mathematics and formalization, complete and machine-checked — only the Born-martingale lemma below carries an open physical premise.

Informal idea. A *glimmer* is a sequence that is random from inside an effective description and structured from one level above it. The paradigm example is Chaitin's Ω [Chaitin, 1975]: no computable betting strategy (martingale) beats it, yet it is fixed by a finite formula and is computable *from* the halting set — it sits at Turing degree $\mathbf{0}'$, i.e. it is Δ_2^0 . (Ω is in fact Martin-Löf random, a strictly stronger property than immunity to merely computable martingales; the formal definition below deliberately uses the weaker *computably random* notion, since that is exactly what the Born-martingale lemma's argument, stated in terms of computable martingales, is capable of establishing — see the derivation chain below. Since Martin-Löf randomness implies computable randomness, Ω satisfies the weaker notion too, so the flagship example is unaffected by this choice.) The claim under investigation, not proved here, is that quantum measurement/adjudication outcomes are glimmers relative to the physical law: not brute randomness, but structure the law itself cannot parse from the inside.

Idealization caveat. Randomness notions — computable randomness, Martin-Löf randomness — are properties of *infinite* sequences. Nothing here commits the paper to an infinite universe: in a finite-information, finite-lifetime universe every realized transcript is finite, and the correct finitary form of the claim below is *prefix incompressibility*, $K(f \upharpoonright n) \geq n - c$ for the realized prefixes relative to the law, for a constant c independent of n . We state the infinite-sequence form throughout for

continuity with the algorithmic-randomness literature cited below; the finitary form is the one that actually applies to any physically realized, finite-lifetime transcript.

Four-class taxonomy. Fix an effective description $E = (\text{computable law with computable Born measure } \mu, \text{ internal observer})$ and let $f : \mathbb{N} \rightarrow \mathbb{N}$ be the adjudication transcript. The four classes below exhaust all possible adjudication transcripts relative to E *conditionally* on a second premise, disclosed in full at derivation-chain step 3 below with the same honesty as the Born-martingale lemma (step 2): UGP Paper 51’s bound [Spivack, 2026] proves $f \leq_T \mathbf{0}'$ for the GTE adjudication layer specifically, under that paper’s own named premise bundle (the Arithmetic Self-Reference record theory and its D_1 – D_5 DSAC/[D]-adjudicator constraints), not as a proved consequence for an arbitrary framework’s transcripts. Ruling out the residual fifth case — non-computable, not computably random, and not Δ_2^0 — for *this* paper’s framework-agnostic **World** model therefore requires importing that bound across a framework boundary; step 3 names this precisely and discloses it as open.

Class	Definition	Example	Status
1. Pseudorandom	f computable	any PRNG	excluded by the non-degeneracy premise [Spivack, 2025d] plus the Born-martingale lemma (below); <i>not</i> by Paper 11 alone
2'. Unpredictable, compressible	f not computable, not computably random, $f \leq_T \mathbf{0}'$	the halting set itself	excluded by the <i>same</i> lemma as class 1 — see the derivation chain below
2. Glimmer	f computably random relative to μ and $f \leq_T \mathbf{0}'$ (i.e. Δ_2^0)	Ω	the claimed class for physical randomness
3. Truly random	f computably random relative to μ , not Δ_2^0	a.e. Martin-Löf-random sequence	excluded by UGP Paper 51 [Spivack, 2026], conditional on the framework-transfer premise (derivation chain, step 3)

Class 2 is measure-zero but nonempty. The distinction between *unpredictable* (no total computable predictor) and *incompressible* (immune to every computable martingale) is exactly what separates class 1 from class 2'; NEMS Paper 11 (“Physical Incompleteness” [Spivack, 2025c]) is cited only as context for this distinction, not as doing the class-1-exclusion work itself — that work is done by the derivation chain that follows.

Derivation chain. Class 1 and class 2' exclusion does not route through Paper 11, nor through any premise about the realized transcript being non-computable as a *conclusion* of a law-level result; instead it falls out of the Born-martingale lemma itself, with Paper 12 supplying a premise rather than a conclusion.

1. **Premise (non-degeneracy / genuine choice).** $\mu(\text{next symbol}) \leq 1 - \varepsilon$ infinitely often along f , for some fixed $\varepsilon > 0$. This is exactly NEMS Paper 12’s own premise, “record-divergent choice” [Spivack, 2025d], so Paper 12 is cited *for the premise*, not for a claim about the transcript’s computability. A degenerate μ (all conditional probabilities eventually 0 or 1) gives no randomness and the question does not arise.
2. **Born-martingale lemma (open).** If μ is complete internal semantics per Papers 13/14 [Spivack, 2025e,f], no computable μ -martingale succeeds on f — i.e. f is computably random

relative to μ . Combined with the elementary fact that a computable, non-degenerate f admits a computable martingale that bets everything on $f(t)$ and succeeds (`Substrate.computable_not_random`, module `Substrate/Glimmer.lean`), this gives $f \notin \text{class 1}$ and $f \notin \text{class 2'}$ simultaneously, since both classes require f to fail computable randomness. **The completeness premise itself — that the physical Born measure genuinely constitutes complete internal semantics, hence that no computable martingale succeeds on the actually realized transcript — is not proved anywhere in the programme;** the gap is narrow and precisely located, not generic. It is *not* a randomness-theory gap: the martingale and computable-randomness machinery the lemma needs is already fully built and machine-checked (section 8.1). Nor can it be closed by simply *defining* “complete internal semantics” to mean “ f is computably random” — that would presuppose exactly what the lemma is meant to establish, foreclosed by this development’s no-smuggling convention; indeed the unrestricted universal form of the premise (every non-degenerate f admits no successful computable martingale) is outright *false* here, since `Substrate.BetMartingale.bet_succeeds` proves the opposite for every *computable* non-degenerate f , so the lemma can only ever concern the specific, presumptively non-computable, realized transcript of a genuinely complete-semantics universe. Nor is a suitable bridge available from the existing quantum-formalism side: the nearest existing candidate for “complete internal semantics,” BICS in the companion `nems-lean` development (Papers 13/14), is a semantic-architecture predicate over density operators and Born-trace probabilities, with no notion of a realized $\mathbb{N} \rightarrow \text{Fin } k$ sequence or computable martingale at all — categorically the wrong kind of object to bridge directly. What remains open is therefore exactly and only a non-question-begging formalization of the physical premise itself as a predicate on realized transcripts; this is the one piece that cannot be closed without new physical or interpretive input, and this paper states the dependency honestly rather than treating it as established.

3. **UGP Paper 51 (adjudication at degree exactly $0'$) — and the framework-transfer premise (open).** $f \leq_T 0' \Rightarrow f \notin \text{class 3}$. UGP Paper 51 [Spivack, 2026] proves this bound for the adjudication layer of the GTE physical framework specifically: its own premise bundle (named PR1–PR5 there) is stated over that framework’s Arithmetic Self-Reference record theory — finite PSC-admissible winding sectors, the D_1 – D_5 constraints on a DSAC/ $[D]$ -class adjudicator — and that paper states explicitly that its theorems are “the GTE-specific physical realization” of a more general, framework-independent claim it attributes to the NEMS formal theory of transputation, not a proof of that general claim itself. This paper’s `World` structure (definition 4.1) is deliberately framework-agnostic (an arbitrary state type, step function, and finite readout; see also the modeling note below) and carries none of that record-theoretic structure, so using $f \leq_T 0'$ for *this* paper’s adjudication transcripts is not a proved consequence of [Spivack, 2026] but a further, currently undischarged **framework-transfer premise**: that whatever process generates a genuinely complete-semantics transcript in this framework instantiates UGP Paper 51’s DSAC/ $[D]$ -class adjudicator model (or the more general abstract theory that paper cites but does not itself prove). No counterexample is known and we are not aware of an argument that the premise fails, but it is logically independent of the Born-martingale lemma above, it is not established by [Spivack, 2026], and the class-3 exclusion here, and the four-class exhaustiveness claim stated at the start of this subsection, should both be read as conditional on it.
4. Hence, conditionally on step 2’s premise and step 3’s framework-transfer premise, $f \in \text{class 2}$ (glimmer).

We note explicitly that step 2’s supporting fact (`Substrate.computable_not_random`) establishes only computable-randomness immunity (against computable martingales), not the strictly stronger Martin-Löf randomness (which requires immunity to the larger class of left-c.e. lower-semicomputable supermartingales); this is exactly why the taxonomy above is stated in terms of computable randomness rather than Martin-Löf randomness — using the stronger notion here would require a correspondingly stronger, less obviously justified premise about what the Born measure’s completeness rules out.

Lean status — A_LEAN, machine-checked. A companion Lean module, `Substrate/Glimmer.lean`, proves exactly the definitions and theorem statements named above (`Substrate.LimitComputable`, `Substrate.BornMeasure`, `Substrate.BornMeasure.NonDegenerateAlong`, `Substrate.BornMeasure.CMartingale`, `Substrate.Succeeds`, `Substrate.ComputablyRandom`, `Substrate.Glimmer`, `Substrate.limitComputable_dense`, `Substrate.notLimitComputable_dense`, `Substrate.limitComputable_not_finitely_determined`, `Substrate.glimmer_finite_description`, `Substrate.computable_not_random`, `Substrate.glimmer_not_computable`, `Substrate.glimmer_disjoint_D`, `Substrate.glimmer_subset_Cstrict`), and is part of this repository’s audited, sorry-free build (imported by `Substrate.lean`; `lake build`: 8262/8262 jobs, zero errors). The bet-everything martingale construction (`Substrate.BetMartingale.bet` / `Substrate.BetMartingale.bet_succeeds`) is fully proved, with no sorry placeholders, via an explicit “wrong-child” dead-branch mechanism handling off-path martingale fairness; `Substrate.glimmer_finite_description` states the non-vacuous fact that a limit-computable sequence is fixed by one code plus a convergent stage function ($(\exists c, g, \text{Computable}(g) \wedge (\forall t, \exists s_0, \forall s \geq s_0, g(t, s) = f(t)) \wedge \text{eval}(c) = \dots)$), not merely an unconditional witness. Every declaration named above is confirmed by direct `#print axioms` to depend on at most the three standard Mathlib axioms (`propext`, `Classical.choice`, `Quot.sound`); see Appendix A for the full table. Non-emptiness of Glimmer for the uniform measure (Chaitin’s Ω) is cited, not built, per Downey–Hirschfeldt [Downey and Hirschfeldt, 2010] — status A_LEAN/D, unaffected by the point above.

Modeling note. The `World` structure (definition 4.1) is deterministic, so a glimmer world is not itself a `World`: this paper’s transcripts are the observer’s channel, and the formal model is deliberately class-agnostic about how the substrate produced them. A glimmer transcript is one *produced by* a deterministic law plus a Δ_2^0 adjudication stream, and Glimmer is defined directly on transcripts, not on `World`. Generalizing `World` with an explicit adjudication input is future work; nothing in Part I needs it.

What it is not. Not a hidden-variable theory: the Δ_2^0 description is of the whole sequence, not a local pre-assignment, so Bell’s theorem [Bell, 1964] is untouched. Not internally usable determinism: theorem 2 still applies (class 2 versus class 3 is itself not finitely determined) and theorem 3 still applies (no internal certifier can fire on this distinction either). The bridge to physical adjudication traces, if the Born-martingale lemma is eventually proved, would be follow-up work via NEMS Papers 76/77 [Spivack, 2025n,o].

Usage rule for this paper and its companions. “Glimmer” means class 2 only, always relative to a named E . Unqualified “random” means classes $2 \cup 3$; “truly random” means class 3 only; “unpredictable” means $\notin$ class 1 and must not be conflated with “random.”

9 Discussion

9.1 Methodological consequence: hunt exponents, not thresholds

The central methodological claim of this paper is that “test whether the substrate is discrete” is not merely a hard experiment to design — it is, for any test of the finitely determined shape, structurally the wrong kind of question, because theorem 2 proves no such test can in principle succeed, independent of instrumental precision or ingenuity. This reframes what counts as progress. A negative result from a Planck-scale interferometry search, or from a search for Lorentz-violating dispersion, is not evidence *against* discreteness in the way it would naively be read; it is evidence against *one particular discretization scheme*, exactly as the causal-set programme demonstrated (section 2). The correct empirical strategy, per section 7, is to hunt *asymptotic scaling exponents* — metrological precision scaling, recurrence-time scaling, resource-versus-instance-size scaling for candidate hypercomputational demonstrations — because these are the one class of observable that is not finitely determined and therefore is not excluded by theorem 2. This is a genuine redirection of experimental priority, not a weakening of empirical ambition: it says the interesting physics is in the exponent, not in the threshold.

9.2 What is proved versus what is cited: the A_Lean / A_MDL line

It matters, and we state it here without hedging, exactly where the machine-checked boundary sits. Lemmas 2.1–2.2, Theorems 1–4, Corollary 4.1, Theorem 4’s Π_2^0 -hardness sharpening (`Substrate.law_match_pi2_hard`), the outsourcing half of Theorem 5 (theorem 5), its uncountable-observer strengthening (corollary 5.1), the deterministic half of the probabilistic-observers result (`Substrate.acceptance_on_D_iff_Cstrict`), the measure-zero/full-measure classification of D/Cstrict (corollary 5.2), the glimmer corollary’s definitions and separations (section 8.1), and the axis-A dense pair (definition 3.1) are all proved and machine-checked (A_Lean, Appendix B) in Lean 4 against Mathlib, with zero `sorry` and zero custom axioms. The only exceptions on the Lean side of the ledger are the Born-martingale lemma’s physical completeness premise and the framework-transfer premise for UGP Paper 51’s degree bound (both section 8.1, genuinely open, not a formalization gap) and the ε -tolerant randomized generalization of the probabilistic-observers result (section 10). The *uniqueness-of-MDL* half of Theorem 5 (theorem 5) is A_MDL: an argument cited to NEMS Papers 34 and 84 [Spivack, 2025l,q], not a Mathlib reduction carried out in this repository. This is not a gap we are trying to paper over — it reflects a genuine difference in the kind of claim being made. “No internally decidable property separates the two transcript classes” is a closed mathematical statement about a formally defined class of procedures, and Lean can check it end to end. “MDL is the unique selection procedure meeting two stated closure desiderata” is itself a theorem, proved elsewhere in the NEMS programme, but importing that proof’s premises (a formal notion of “selection procedure invariant under presentation”) into this repository’s vocabulary is future formalization work, not a philosophical shortcut. We flag this explicitly so that no reader mistakes the paper’s overall confidence in Theorem 5 for uniform machine-checked confidence in both of its halves.

9.3 Relation to the digital-physics debate

This paper’s stance toward Zuse, Fredkin, Wolfram, and ’t Hooft on one side, and an unrestricted-continuum view on the other, is neutrality of a specific and strong kind: not “we do not know which is right,” but “neither can be internally proved right, and this is a theorem, not an admission of ignorance.” That is a stronger and different claim than declining to take a side in the ordinary

sense. A digital-physics advocate who points to a successful match between a candidate discrete rule and observed physics, however striking the match, is producing exactly the situation theorem 4 addresses: a candidate law that has matched so far, with no internal procedure available to certify that the match will continue forever or that it is not one of the countably many discrete laws that happens to agree with observations up to the current prefix while a continuum, or a different discrete law, diverges downstream. Symmetrically, a continuum advocate who treats general relativity’s differentiable manifold as established fact about the actual substrate, rather than as the best-performing finitely-described theory to date, is claiming a certification that theorem 4 shows is unavailable to them too. The paper’s neutrality is therefore not agnosticism about physics; it is a proof that the entire debate, as usually conducted (each side citing empirical success as if it amounted to internal proof), is arguing for something structurally unprovable from the inside, regardless of which side is factually correct.

9.4 Honest limitations beyond section 6

section 6 states the theorems’ technical limits; here we state programmatic ones. First, if computable analysis is eventually added to Mathlib, the refinement sketched in definition 4.4 and section 6 item 4 should be carried out as a genuine formalization project, not merely asserted via Pour-El–Richards; until then, the v1 development’s reliance on an unrestricted classical continuum is a real, disclosed simplification, not a hidden one. Second, the quantum extension sketched in section 10 — replacing **World** with a finite-dimensional versus separable Hilbert space and a POVM readout — is currently only a sketch: theorem 1 is expected to go through essentially unchanged, but whether theorem 4 survives with a unitary step is a genuinely open question, since the reduction’s diagonal construction currently exploits classical branching (**if-then-else** on a halting check) that does not obviously transfer to unitary evolution without decoherence or measurement entering the model. Third, axis C (objective randomness) is addressed directly: the glimmer corollary (section 8.1) is proved in this paper, with its definitions and derivation chain machine-checked, because theorem 5’s conclusion is not correctly statable without it. Two honest gaps remain, and we disclose both precisely rather than rounding them up to a claimed result: the Born-martingale lemma itself — that the actual physical Born measure admits no successful computable martingale on the realized transcript — is not yet proved anywhere in the programme; and the framework-transfer premise — that UGP Paper 51’s degree-0’ bound, proved there for the GTE adjudication layer under that paper’s own premise bundle, transfers to this paper’s framework-agnostic **World** model — is likewise not proved anywhere. Those are the *only* gaps: the Lean module proving the classification’s mathematical content (`Substrate/Glimmer.lean`) is part of this repository’s audited, sorry-free build, as disclosed in full in section 8.1.

9.5 Forward-looking directions

Each remaining item in section 10 matters for a distinct reason, stated there in full; in summary: the deterministic half of the probabilistic-observers question is closed (section 10), and Baire space carries a genuine probability measure under which discreteness is measure-zero-typical (corollary 5.2); what remains is genuinely closer to real experimental practice — a *randomized* test with some advantage short of certainty, evaluated against a measure that does not already assign full mass to one class by construction. The cardinality strengthening of Theorem 5, corollary 5.1, forecloses a natural “pool enough observers” escape attempt, showing the barrier survives even an uncountable family of cooperating internal observers. And the glimmer corollary’s missing Born-martingale lemma, together with the framework-transfer premise for UGP Paper 51’s degree bound, are the two

results standing between this paper’s programme and a genuine resolution of axis C’s relationship to axis B, which is why we name them explicitly as the actual formal target rather than leaving axis C as an unstructured gap.

10 Open Problems and Extensions

- **Computable-analysis refinement of ContinuumWorld.** Restrict step to Cauchy-modulus-computable maps on Cauchy-modulus-computable reals. `Cstrict` provably remains nonempty under this refinement, by Pour-El–Richards [Pour-El and Richards, 1981] (section 2, with the regularity caveat noted there). Precisely what is missing: current Mathlib provides no computable-analysis framework at all — no notion of a Cauchy-modulus-computable real, no Weihrauch-degree machinery, nothing analogous to the `Computable` predicate on $\mathbb{N} \rightarrow \mathbb{N}$ already used throughout this development (verified by direct search of the Mathlib source: zero occurrences of `Weihrauch` or `CauchyModulus` anywhere in the library as of this writing). Formalizing this refinement therefore means building a computable-analysis layer largely from scratch, not applying existing machinery; that is well beyond this paper’s scope.
- **Quantum version.** Replace `World` with a finite-dimensional versus separable Hilbert space and a POVM readout. theorem 1 is expected to go through with essentially the same proof strategy. Whether theorem 4 survives with a unitary step is open (section 9). Precisely what is missing: Mathlib currently has zero formalization of POVMs, density operators, or completely-positive trace-preserving (CPTP) maps (verified by direct search), so even *stating* the quantum extension formally — not merely proving theorem 4’s quantum analogue — is future infrastructure work, not a proof gap in the existing development.
- **The glimmer corollary’s two open premises** (section 8.1) — the definitions, four-class taxonomy, and derivation chain of the glimmer corollary are settled Part II content (section 8.1), not an open problem; what remains open are two distinct physical/interpretive premises. The **Born-martingale lemma**: that the actual physical Born measure, as complete internal semantics per NEMS Papers 13/14 [Spivack, 2025e,f], admits no successful computable martingale on the realized transcript. This is not a randomness-theory gap — the martingale and computable-randomness machinery the lemma needs is already fully built and machine-checked (section 8.1) — and it cannot be closed by definitional fiat without smuggling the conclusion. The **framework-transfer premise**: that UGP Paper 51’s degree-0’ bound [Spivack, 2026] — proved there for the GTE adjudication layer under that paper’s own named premise bundle, which is stated over that framework’s specific record theory and not for an arbitrary framework’s transcripts — transfers to this paper’s framework-agnostic `World` model, which shares none of that bundle’s record-theoretic structure. Neither premise is discharged by this paper’s own machinery, and they are logically independent of one another; see section 8.1 for the full derivation chain and precisely why these two premises, and only these two, remain open.
- **Probabilistic observers.** The deterministic class-advantage obstruction is proved: an internal test has positive acceptance on `D` if and only if it has positive acceptance on `Cstrict` (`Substrate.acceptance_on_D_iff_Cstrict`, module `Substrate/Probabilistic.lean`), so no internal test exhibits exclusive class advantage (`Substrate.not_exclusive_acceptance`) — the prefix-lifting of theorem 2 from properties to a test’s acceptance behavior, by the identical mechanism (theorem 1). Baire space carries a genuine measure-theoretic treatment: `Substrate/Probabilistic.lean` wires in Mathlib’s Ionescu–Tulcea-based

`MeasureTheory.Measure.infinitePi` directly onto $\mathbb{N} \rightarrow \mathbb{N}$, giving a canonical geometric i.i.d. product measure under which `D` is null and `Cstrict` has full measure (corollary 5.2). What remains genuinely open is a *genuinely randomized*, error-tolerant test. Prefix indistinguishability (theorem 1) transports individual witnesses between classes, but it does not bound *aggregate* class-conditional acceptance-rate differences under a sampling measure: the distribution of prefixes under a `D`-conditioned sample and a `Cstrict`-conditioned sample can differ arbitrarily, so witness-transport alone gives no rate bound. A non-trivial ε -tolerant result requires a genuinely randomized test — one with independent internal randomness, not merely a prefix-determined confidence function — evaluated against a *mixture* measure that assigns controllable positive mass to both `D` and `Cstrict` separately, since under any single i.i.d. product measure on the full transcript space `D` is null and `Cstrict` has full mass (corollary 5.2), so conditioning on `D` alone is degenerate under that measure. A witness-based restatement using a prefix-determined confidence function (`Substrate.HasProbAcceptAbove`) turns out to add no independent content on inspection: since the confidence value depends only on the prefix, theorem 1 transports *identical* confidence values between classes, not merely ε -close ones, so the tolerance parameter never does any real work in that construction — it is a notational restatement of `Substrate.acceptance_on_D_iff_Cstrict`, not an independent strengthening, and is omitted from Appendix A on those grounds.

11 Conclusion

We have shown, and machine-checked in Lean 4 against Mathlib with zero `sorry` and zero custom axioms, that no finite-resource observer internal to a physical system can decide whether that system’s substrate is discrete or continuum, on either of the two readings that matter (information density and dynamics). The obstruction comes in two independent forms: a topological barrier that needs no computability theory at all, and a recursion-theoretic barrier that reduces from the halting problem even when the observer is handed a complete candidate law. Under Perfect Self-Containment, this makes substrate discreteness an outsourced property, and the only lawful internal-consistent adjudicator — by an argument cited from the companion NEMS programme, not re-derived here (section 9) — is minimum-description-length selection over complete law-sets: not a methodological preference, but the unique procedure that survives the closure requirement. This dissolves the traditional grid-search experimental programme’s claim to settle the question, redirects empirical attention to asymptotic scaling exponents, and places the epistemic status of finite, compressive physical theories — including those coming out of the UGP programme — exactly where Theorem 5 puts it: strong evidence, never certification. The results are not silent on objective randomness: the glimmer corollary’s definitions and separations are machine-checked (section 8.1) — but the Born-martingale lemma’s physical completeness premise, the framework-transfer premise connecting UGP Paper 51’s degree-0’ bound to this paper’s framework-agnostic model, and the question of which substrate actually obtains, are left, honestly, as open.

A Lean Statement List

Axiom-printout status: **final.** `Substrate/*.lean` builds cleanly (lake build: 8262/8262 jobs, zero errors) and `grep -r sorry Substrate/` finds nothing. The `#print axioms` column below was obtained by running `lake env lean scripts/axiom_audit.lean` (the declarations that script covers include every `Substrate/Glimmer.lean` and `Substrate/AxisA.lean` declaration named in section 8.1

and definition 3.1, plus `Substrate.law_match_pi2_hard` and `Substrate.law_match_pi2_hard_zero`) and, for the remaining declarations in this table, an equivalent direct `#print axioms` invocation against the built library, both re-run independently for this draft. Every named theorem depends on at most the three standard Mathlib axioms (`propext`, `Classical.choice`, `Quot.sound`); one (`Substrate.finitely_determined_compl`) depends on *no* axioms at all. Zero custom axioms, confirmed.

Lean declaration	Statement	#print axioms
<code>Substrate.transcript_characterisation</code>	Lemma 2.1: $f \in D \leftrightarrow \text{Computable } f$	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.discrete_subset_continuum</code>	Lemma 2.2: $D \subseteq C$	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.mem_C</code>	Every $f : \mathbb{N} \rightarrow \mathbb{N}$ lies in C	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.exists_noncomputable_nat_fn</code>	$\exists f, \neg \text{Computable } f$	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.not_computable_halting_Indicator</code>	Halting-indicator witness is not computable	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.prefix_indistinguishable</code>	Theorem 1: every prefix extends into both D and C_{strict}	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.D_disjoint_Cstrict</code>	D and C_{strict} disjoint	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.not_finitely_determined_of_dense_pair</code>	Abstract dense-pair form of Theorem 2, for arbitrary disjoint prefix-dense $A, B \subseteq \mathbb{N} \rightarrow \mathbb{N}$; no <code>Computable/Nat.Partrec</code> dependence in the proof term	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.substrate_not_finitely_determined</code>	Theorem 2: no finitely determined property separates D from C_{strict} , as the D/C_{strict} instance of the abstract lemma above	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.sound_for_continuum_never_fires</code>	Theorem 3(a): continuum-sound test never fires	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.sound_for_discreteness_never_fires</code>	Theorem 3(b): discreteness-sound test never fires	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.certifier_silent_or_unsound</code>	Theorem 3 combined	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.law_match_undecidable</code>	Theorem 4: law-match predicate is not computable, for any computable f	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.law_match_undecidable_zero</code>	Corollary 4.1: Theorem 4 specialized to $f \equiv 0$ (Paper 30 bridge)	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>

continued on next page

Lean declaration	Statement	#print axioms
Substrate.law_match_pi2_hard	Theorem 4': $\text{Tot} \leq_m \text{LawMatches}(f, \cdot)$ for any computable f , via a sequential-composition reduction (run the candidate to completion, discard the result, emit f); Tot's Π_2^0 -completeness is external, not proved here	propext, Classical.choice, Quot.sound
Substrate.law_match_pi2_hard_zero	Theorem 4' specialized to $f \equiv 0$	propext, Classical.choice, Quot.sound
Substrate.internally_decidable_implies_finitely_determined	Internally decidable $\Rightarrow$ finitely determined	propext, Classical.choice, Quot.sound
Substrate.finitely_determined_compl	Complement of a finitely determined property is finitely determined	none
Substrate.substrate_not_internally_decidable	Theorem 5 (Lean half): no internally decidable property separates D from Cstrict	propext, Classical.choice, Quot.sound
Substrate.substrate_not_internally_decidable'	Symmetric separation	propext, Classical.choice, Quot.sound
Substrate.discrete_not_internally_decidable	$f \in D$ not internally decidable	propext, Classical.choice, Quot.sound
Substrate.strict_continuum_not_internally_decidable	$f \in \mathbf{Cstrict}$ not internally decidable	propext, Classical.choice, Quot.sound
Substrate.D_countable	D is countable	propext, Classical.choice, Quot.sound
Substrate.Cstrict_not_countable	Cstrict is not countable	propext, Classical.choice, Quot.sound
Substrate.no_joint_substrate_decision	corollary 5.1: no family of internal tests, over any index type, jointly decides discrete membership	propext, Classical.choice, Quot.sound
Substrate.acceptance_on_D_iff_Cstrict	Deterministic class-advantage obstruction: positive acceptance on D iff positive acceptance on Cstrict	propext, Classical.choice, Quot.sound
Substrate.D_null_geometric	corollary 5.2: D has measure zero under the geometric i.i.d. product on Baire space	propext, Classical.choice, Quot.sound
Substrate.Cstrict_full_mass_geometric	corollary 5.2: Cstrict has full measure under the same measure	propext, Classical.choice, Quot.sound
Substrate.limit_Computable_not_finitely_determined	section 8.1: the abstract dense-pair form of Theorem 2 applied to limit-computable vs. not-limit-computable sequences	propext, Classical.choice, Quot.sound
Substrate.computable_not_random	section 8.1, derivation-chain step 2's supporting fact: a computable, non-degenerate f admits a computable μ -martingale that succeeds, hence is not computably random	propext, Classical.choice, Quot.sound
Substrate.glimmer_not_computable	section 8.1: a glimmer sequence is not computable	propext, Classical.choice, Quot.sound
Substrate.glimmer_disjoint_D	section 8.1: Glimmer is disjoint from D	propext, Classical.choice, Quot.sound

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Lean declaration	Statement	#print axioms
<code>Substrate.glimmer_subset_Cstrict</code>	section 8.1: every glimmer sequence lies in <code>Cstrict</code>	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.glimmer_finite_description</code>	section 8.1: a limit-computable sequence is fixed by one code plus a convergent stage function (non-vacuous finite-description witness)	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.BetMartingale.bet_succeeds</code>	section 8.1: the bet-everything martingale construction succeeds on any computable, non-degenerate f	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.Dfin_iff_eventuallyPeriodic</code>	definition 3.1: <code>Dfin</code> coincides with the eventually periodic sequences	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>Substrate.axisA_not_finitely_determined</code>	definition 3.1: the abstract dense-pair form of Theorem 2 applied to <code>Dfin</code> vs. its complement	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>

Module map.

Module	Role
<code>Substrate/Basic.lean</code>	<code>Substrate.World</code> , transcript, <code>Substrate.D</code> / <code>Substrate.C</code> / <code>Substrate.Cstrict</code> , <code>Substrate.InternalTest</code> , <code>Substrate.FinitelyDetermined</code>
<code>Substrate/Discrete.lean</code>	Lemma 2.1
<code>Substrate/Embed.lean</code>	Lemma 2.2
<code>Substrate/Noncomputable.lean</code>	Existence of a non-computable $\mathbb{N} \rightarrow \mathbb{N}$ function, via the halting problem
<code>Substrate/Prefix.lean</code>	Theorem 1 and the disjointness/density corollary
<code>Substrate/NotFinDet.lean</code>	Theorem 2, and its abstract dense-pair generalization (theorem 2's keybox)
<code>Substrate/Certifier.lean</code>	Theorem 3
<code>Substrate/LawBarrier.lean</code>	Theorem 4 (halting reduction via <code>evaln</code>) and Theorem 4' (Π_2^0 -hardness via the sequential-composition reduction)
<code>Substrate/Outsourced.lean</code>	Theorem 5 (Lean-checkable half)
<code>Substrate/Cardinality.lean</code>	corollary 5.1: countability of <code>D</code> , non-countability of <code>Cstrict</code> , and the uncountable-observer strengthening
<code>Substrate/Probabilistic.lean</code>	Deterministic class-advantage obstruction for probabilistic observers, and measure-zero / full-measure classification of <code>D</code> / <code>Cstrict</code> under a geometric product measure (corollary 5.2 and section 10)
<code>Substrate/Glimmer.lean</code>	section 8.1: limit-computability, computable Born measures, computable martingales, computable randomness, and the glimmer class, with the bet-everything martingale construction and the class-1/2'/2/3 separation theorems
<code>Substrate/AxisA.lean</code>	definition 3.1: the axis-A dense pair <code>Dfin</code> (eventually periodic sequences) and its own instance of theorem 2

B Epistemic Status Table

Claim	Status
Lemma 2.1, Lemma 2.2; Theorems 1–4; Corollary 1.1, Corollary 4.1	A_LEAN (machine-checked, Lean 4 / Mathlib, zero <code>sorry</code> , zero custom axioms)
Abstract dense-pair form of Theorem 2 (theorem 2’s keybox)	A_LEAN (module <code>Substrate/NotFinDet.lean</code> ; no <code>Computable/Nat.Partrec</code> dependence in the proof term)
Theorem 4’ (Π_2^0 -hardness of <code>LawMatches</code> , <code>Substrate.law_match_pi2_hard/Substrate.law_match_pi2_hard_zero</code>)	A_LEAN (module <code>Substrate/LawBarrier.lean</code> ; $\text{Tot} \leq_m \text{LawMatches}(f, \cdot)$ via a sequential-composition reduction; Tot’s Π_2^0 -completeness is an external, cited recursion-theoretic fact, not proved in this repository)
Theorem 5, outsourcing half (theorem 5) Theorem 5, MDL-uniqueness half (theorem 5)	A_LEAN A_MDL (argument cited to NEMS Papers 34/84; not a Mathlib reduction)
Uncountable-observer strengthening of Theorem 5 (corollary 5.1)	A_LEAN (module <code>Substrate/Cardinality.lean</code> ; closed no bounds on the index type I , including uncountable)
Probabilistic observers, deterministic class-advantage obstruction (<code>Substrate.acceptance_on_D_iff_Cstrict</code>)	A_LEAN (module <code>Substrate/Probabilistic.lean</code>)
Probabilistic observers, measure-zero / full-measure classification of D / <code>Cstrict</code> under a geometric product measure (corollary 5.2)	A_LEAN (module <code>Substrate/Probabilistic.lean</code> ; uses Mathlib’s Ionescu–Tulcea-based <code>Measure.infinitePi</code>)
Probabilistic observers, ε -tolerant <i>randomized</i> generalization (genuine internal randomness plus a class-charged mixture measure)	Open — prefix indistinguishability transports individual witnesses but not aggregate class-conditional acceptance rates; a prefix-determined witness restatement (<code>Substrate.HasProbAcceptAbove</code>) adds no independent content and is omitted from Appendix A on those grounds (section 10)
§7 scaling-exponent classification	D (discussion-level, not machine-checked, presented with citations)

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Claim	Status
§8.1 glimmer definitions (<code>Substrate.LimitComputable</code>, <code>Substrate.BornMeasure</code>, <code>Substrate.Glimmer</code>, etc.), limit-computability density pair, <code>Substrate.limitComputable_not_finitely_determined</code>, <code>Substrate.glimmer_finite_description</code>, <code>Substrate.computable_not_random</code>, the bet-everything martingale construction (<code>Substrate.BetMartingale.bet</code>/<code>Substrate.BetMartingale.bet_succeeds</code>), and the separations built on it (<code>Substrate.glimmer_not_computable</code>, <code>Substrate.glimmer_disjoint_D</code>, <code>Substrate.glimmer_subset_Cstrict</code>) §8.1 Glimmer non-emptiness (Chaitin’s Ω)	A <code>_LEAN</code> (module <code>Substrate/Glimmer.lean</code>, part of the audited, sorry-free build; imported by <code>Substrate.lean</code>; every declaration confirmed by <code>#print axioms</code> to depend on only <code>propext</code>, <code>Classical.choice</code>, <code>Quot.sound</code>) A <code>_LEAN/D</code> (cited to Downey–Hirschfeldt [Downey and Hirschfeldt, 2010], not built, by design)
§8.1 Born-martingale lemma (the physical completeness premise)	Open — not proved here or anywhere in the programme at time of writing. Not a randomness-theory gap: the martingale/computable-randomness machinery (<code>Substrate.computable_not_random</code>, <code>Substrate.BetMartingale.bet_succeeds</code>) is machine-checked (row above); the unrestricted universal form of the premise is in fact <i>false</i> here (<code>Substrate.BetMartingale.bet_succeeds</code> beats every <i>computable</i> non-degenerate f), so defining “complete internal semantics” as “f is computably random” would smuggle the conclusion. The nearest existing candidate, <code>nems-lean</code>’s BICS (Papers 13/14), is a quantum semantic-architecture predicate over density operators, categorically the wrong kind of object for a realized-sequence bridge. What is missing is specifically a non-question-begging formalization of the physical premise as a predicate on realized transcripts (section 10)

continued on next page

Claim	Status
§8.1 framework-transfer premise (UGP Paper 51’s degree- $\mathbf{0}'$ bound applied to this paper’s transcripts)	Open — not proved here or anywhere in the programme at time of writing, and logically independent of the Born-martingale lemma (row above). UGP Paper 51 [Spivack, 2026] proves $f \leq_T \mathbf{0}'$ for the GTE adjudication layer specifically, under that paper’s own named premise bundle (its Arithmetic Self-Reference record theory and D_1 – D_5 DSAC/[D]-adjudicator constraints), and that paper itself attributes the framework-independent form of the claim to a separate work (the NEMS formal theory of transputation), not to its own theorems. This paper’s World structure (definition 4.1) is deliberately framework-agnostic and carries none of that record-theoretic structure, so importing the bound for <i>this</i> paper’s transcripts is a further, undischarged premise, not a proved consequence of [Spivack, 2026]
definition 3.1 axis-A pair (<code>Substrate.Dfin</code> , <code>Substrate.Dfin_iff_eventuallyPeriodic</code> , <code>Substrate.axisA_not_finitely_determined</code>)	<code>A_LEAN</code> (module <code>Substrate/AxisA.lean</code> , part of the audited, sorry-free build; imported by <code>Substrate.lean</code> ; every declaration confirmed by <code>#print axioms</code> to depend on only <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>)

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